

Simulation of basic semiconductor devices

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This document explains the methods and equations which can be used for numerical solution of electrical characteristics of basic semiconductor devices. MATLAB codes based on these are available for download from the same directory. The codes are written for educational purpose. The students can play with the codes and understand how exactly numerical solutions to basic semiconductor equations are obtained. Some of the simulations and derivations of few equations are left as exercise.

These codes can be used to calculate the following:

- 1) pn junction band diagram in equilibrium (works for hetero-junctions, non-degenerate doping as well. Can use iterative or non-iterative method.)
- 2) DC IV of pn junctions (homojunctions and non-degenerate doping)
- 3) Small signal admittance (frequency and bias dependent) of pn junctions (homojunctions and non-degenerate doping)
- 4) Small signal admittance (frequency and bias dependent) of ideal MOS. (works for non-degenerate doping as well)
- 5) Simulation of Haynes-Shockley experiment (Time dependent simulation)

All the above codes are based on 1d models. SRH (Shockley–Read–Hall) model has been used for recombination-generation processes. Electric field dependent mobility is included. In most of the cases, the simulated characteristics are compared with what is expected from the theory.

The codes are written in MATLAB. They use ode45 (for solving differential equation), fzero (for finding zero of a function) and back slash operator (for solution of algebraic equations) of MATLAB. These can be replaced by your own functions such as Runge Kutta method for solving differential equations, Newton-Raphson method to find zero of a function and LU decomposition method to solve algebraic equations.

It is assumed that the reader has taken a course on semiconductor devices and is familiar with the basics of semiconductors, pn junctions, depletion approximation, energy band diagrams, MOS capacitor, MOSFET etc.

The document has the following sections:

1. Basic equations for uniformly doped semiconductor in equilibrium:
2. Equilibrium band diagram of pn junction:
 - A. Direct method to solve Poisson equation
 - B. Iterative method to solve Poisson equation

3. Calculation of dc current as a function of dc voltage in pn junction
4. Calculation of small signal admittance of pn junction
5. Calculation of capacitance of MOS-capacitor
6. Time dependent simulations

A: Simulation of Haynes-Shockley experiment

Appendix A: Derivation of Scharfetter-Gummel discretization Scheme:

Appendix B: Equations for Degenerate case

Appendix C: Discretization of electron and hole current densities in the degenerate case

Appendix D: Kramer-Kronig relations

References

1. Basic equations for a uniformly doped semiconductor in equilibrium:

The basic equations for a uniformly doped semiconductor in equilibrium are given below. The approximate equations used in non-degenerate case ($E_C - E_F > 3k_B T$, and $E_F - E_V > 3k_B T$) are also given below (See e.g. Pierret). These equations are used in the simulation program.

$$\left. \begin{aligned} n &= N_c \mathcal{F}_{1/2}\left(\frac{E_F - E_C}{k_B T}\right) \text{ and } p = N_v \mathcal{F}_{1/2}\left(\frac{E_V - E_F}{k_B T}\right), \text{ where } \mathcal{F}_{1/2}(\eta) = \frac{2}{\sqrt{\pi}} \int_0^\infty dx \frac{\sqrt{x}}{1 + \exp(x - \eta)} \\ N_D^+ &= \frac{N_D}{1 + g_D \exp\left[\left(\frac{E_F - E_d}{k_B T}\right)\right]}, \quad N_A^- = \frac{N_A}{1 + g_A \exp\left[\left(\frac{E_a - E_F}{k_B T}\right)\right]} \\ \rho &= q(p - n + N_D^+ - N_A^-) \end{aligned} \right\} \text{--- (1.1)}$$

$$\left. \begin{aligned} &\text{non-degenerate approximation } (E_C - E_F \geq 3k_B T \text{ and } E_F - E_V \geq 3k_B T): \\ n &= N_c \exp\left(\frac{E_F - E_C}{k_B T}\right), \quad p = N_v \exp\left(\frac{E_V - E_F}{k_B T}\right) \\ N_D^+ &= N_D, \quad N_A^- = N_A \text{ (assumed complete ionization.)} \\ n_i &= \sqrt{np} = \sqrt{N_c N_v} \exp\left(\frac{E_V - E_C}{2k_B T}\right) = \sqrt{N_c N_v} \exp\left(\frac{-E_g}{2k_B T}\right) \\ E_i &= \frac{E_C + E_V}{2} + \frac{1}{2} k_B T \ln\left(\frac{N_v}{N_c}\right) \\ E_F &= E_i + k_B T \sinh^{-1}\left(\frac{N_D - N_A}{2n_i}\right) \\ \rho &= q(p - n + N_D - N_A) \end{aligned} \right\} \text{--- (1.2)}$$

Here all the symbols have usual meaning. (Note that complete ionization has been included as a part of non-degenerate approximation.) In the case of pn junction or MOS capacitor, there is band bending near the interface in equilibrium. Denoting the potential measured w.r.t. bulk intrinsic Fermi level by ϕ , all the energy levels in above equations (except E_F) should be replaced with $E \rightarrow E - q\phi(x)/k_B T$ to get n , p , N_D^+ and N_A^- . We get:

$$\left. \begin{array}{l}
n = N_c \mathcal{F}_{1/2} \left(\frac{E_F - E_C + q\phi}{k_B T} \right) \\
p = N_v \mathcal{F}_{1/2} \left(\frac{E_V - q\phi - E_F}{k_B T} \right) \\
N_D^+ = \frac{N_D}{1 + g_D \exp \left[\left(\frac{E_F - E_d + q\phi}{k_B T} \right) \right]} \\
N_A^- = \frac{N_A}{1 + g_A \exp \left[\left(\frac{E_a - q\phi - E_F}{k_B T} \right) \right]} \\
\text{degenerate}
\end{array} \right\} \quad \left. \begin{array}{l}
n = N_c \exp \left(\frac{E_F - E_C + q\phi}{k_B T} \right) \\
p = N_v \exp \left(\frac{E_V - q\phi - E_F}{k_B T} \right) \\
N_D^+ = N_D \\
N_A^- = N_A \\
\text{non-degenerate}
\end{array} \right\} \text{--- (1.3)}$$

Let us now consider a uniformly doped semi-infinite semiconductor (extending from $x=0$ to ∞) in equilibrium. The potential $\phi(x)$, satisfies the Poisson equation:

$$\frac{d^2 \phi}{dx^2} = -\frac{\rho(\phi)}{\epsilon_0 K_S} \text{--- (1.4)}$$

If we multiply the above equation by $d\phi/dx$, the LHS side can be written as

$$LHS = \frac{d^2 \phi}{dx^2} \frac{d\phi}{dx} = \frac{1}{2} \frac{d}{dx} \left(\frac{d\phi}{dx} \right)^2. \text{ If we integrate from } x \text{ to } \infty, \text{ we get:}$$

$$\begin{aligned}
\frac{1}{2} \left[\left(\frac{d\phi}{dx} \right)^2 \right]_x^\infty &= \int_x^\infty -\frac{\rho(\phi)}{\epsilon_0 K_S} \frac{d\phi}{dx} dx' = \int_\phi^0 -\frac{\rho(\phi')}{\epsilon_0 K_S} d\phi' \\
\Rightarrow \frac{1}{2} \left(\frac{d\phi}{dx} \right)^2 &= \frac{-1}{\epsilon_0 K_{Si}} \int_0^\phi \rho(\phi') d\phi', \quad \text{put } \frac{d\phi}{dx} = 0 \text{ at } \infty \\
\Rightarrow E^2 &= \left(\frac{d\phi}{dx} \right)^2 = \frac{-2}{\epsilon_0 K_{Si}} \int_0^\phi \rho(\phi') d\phi' \\
\Rightarrow E &= -\frac{d\phi}{dx} = \text{sign}(\phi) \sqrt{\frac{-2}{\epsilon_0 K_S} \int_0^\phi \rho(\phi') d\phi'} \text{--- (1.5)}
\end{aligned}$$

Note that above equation has a factor of $\text{sign}(\phi)$. This equation holds for the semi-infinite semiconductor lying from $x=0$ to ∞ . For semi-infinite semiconductor lying from $x=0$ to $-\infty$, multiply RHS of above equation by -1.

(exercise: show from above equation that if the potential is small ($\ll k_B T/q$), the potential decays exponentially i.e. $\phi(x) \approx \phi(0) \exp(-x/l)$, with length scale $l = \sqrt{-\epsilon_0 K_{Si} / \left(\frac{d\rho}{d\phi} \right)_0}$. This is called as Debye length.)

Electric field as a function of potential, $E(\phi)$ is a very useful equation. If we know the value of potential at any point x_0 , we can solve the differential equation $d\phi/dx = -E(\phi)$, to get potential

as a function of x . (Since this is a first order equation, we need only one boundary condition.) We then know all quantities such as density of electrons, holes, ionized donors, ionized acceptors, charge density, electric field as a function of x . Using equation 1.5 we will now find out the function $E(\phi)$ for non-degenerate and degenerate cases. (Note that while deriving equation 1.5, we have assumed a semi-infinite semiconductor with electric field $=0$ at $x=\infty$. That is why only one boundary condition is required now to get potential as a function of x .)

A. Getting electric field, $E(\phi)$ for non-degenerate case: We now consider non-degenerate case i.e. $(E_C - q\phi/k_B T - E_F > 3k_B T)$, and $E_F - E_V + q\phi(x)/k_B T > 3k_B T$. In this case, the electron and hole densities can be written as (see non-degenerate block of equation 1.3):

$$p = p_0 \exp\left(-\frac{q\phi}{k_B T}\right), \quad n = n_0 \exp\left(\frac{q\phi}{k_B T}\right) \quad \text{--- (1.6)}$$

where p_0 and n_0 are the hole and electron concentrations in the bulk.

$$n_0 = N_c \exp[(E_F - E_C) / k_B T] \quad \text{and} \quad p_0 = N_v \exp[(E_V - E_F) / k_B T] \quad \text{--- (1.7)}$$

The charge density in the non-degenerate case is given by: $\rho = q(p - n + N_D - N_A)$. Inserting this expression in equation 1.5, and carrying out the integration, we get:

$$E = -\frac{d\phi}{dx} = \text{sign}(\phi) \sqrt{\frac{2k_B T}{\epsilon_0 K_s}} \left\{ p_0 \left[\exp\left(-\frac{q\phi}{k_B T}\right) - 1 \right] + n_0 \left[\exp\left(\frac{q\phi}{k_B T}\right) - 1 \right] + (N_A - N_D) \frac{q\phi}{k_B T} \right\}^{1/2} \quad \text{--- (1.8)}$$

This equation holds for the semiconductor lying from $x=0$ to ∞ . For a semiconductor lying from $x=0$ to $-\infty$, multiply RHS of above equation by -1 .

(Exercise: Show from equation 1.6 that the Debye length is given by, $\sqrt{\epsilon_0 K_s k_B T / q^2 (p_0 + n_0)}$.)

B. Getting electric field, $E(\phi)$ for degenerate case: Here we use the expression for p , n , N_D^+ and N_A^- given by the degenerate block of equation 1.3. We follow the steps given below to get electric field as a function of potential:

$$p = N_v \mathcal{F}_{1/2}\left(\frac{E_V - E_F - q\phi}{k_B T}\right) \\ \Rightarrow \int_0^\phi p(\phi') d\phi' = N_v \int_0^\phi \mathcal{F}_{1/2}\left(\frac{E_V - E_F - q\phi'}{k_B T}\right) d\phi' = \frac{-k_B T}{q} N_v \left[\mathcal{F}_{3/2}\left(\frac{E_V - E_F - q\phi}{k_B T}\right) - \mathcal{F}_{3/2}\left(\frac{E_V - E_F}{k_B T}\right) \right] \quad \text{--- (1.9)}$$

$$n = N_c \mathcal{F}_{1/2}\left(\frac{E_F - E_C + q\phi}{k_B T}\right) \\ \Rightarrow \int_0^\phi n(\phi') d\phi' = N_c \int_0^\phi \mathcal{F}_{1/2}\left(\frac{E_F - E_C + q\phi'}{k_B T}\right) d\phi' = \frac{k_B T}{q} N_c \left[\mathcal{F}_{3/2}\left(\frac{E_F - E_C + q\phi}{k_B T}\right) - \mathcal{F}_{3/2}\left(\frac{E_F - E_C}{k_B T}\right) \right] \quad \text{--- (1.10)}$$

$\mathcal{F}_{3/2}$ in above equations is Fermi-Dirac integral of order $3/2$. In writing above equations, we have used a known relation between the Fermi-Dirac integrals ($j > 0$) viz.

$$\frac{d}{d\eta} \mathcal{F}_j(\eta) = \mathcal{F}_{j-1}(\eta) \quad i.e. \int \mathcal{F}_{j-1}(\eta) d\eta = \mathcal{F}_j(\eta) + c \quad --- (1.11)$$

where j is the order of Fermi-Dirac integral. This can be proved as follows:

$$\left. \begin{aligned} \mathcal{F}_j(\eta) &= \frac{1}{\Gamma(j+1)} \int_0^\infty dx \frac{x^j}{1 + \exp(x - \eta)}, \quad \Gamma = \text{gamma function} \\ \Rightarrow \frac{d}{d\eta} \mathcal{F}_j(\eta) &= \frac{1}{\Gamma(j+1)} \int_0^\infty dx \frac{d}{d\eta} \frac{x^j}{1 + \exp(x - \eta)} \\ \text{use: } \frac{d}{d\eta} \frac{x^j}{1 + \exp(x - \eta)} &= -\frac{d}{dx} \frac{x^j}{1 + \exp(x - \eta)} + j \frac{x^{j-1}}{1 + \exp(x - \eta)} \\ \Rightarrow \frac{d}{d\eta} \mathcal{F}_j(\eta) &= 0 + \frac{1}{\Gamma(j)} \int_0^\infty dx \frac{x^{j-1}}{1 + \exp(x - \eta)} = \mathcal{F}_{j-1}(\eta) \\ \text{used: } \Gamma(j+1) &= j \Gamma(j) \end{aligned} \right\} --- (1.12)$$

We also need integrals of N_D^+ and N_A^- . They are obtained as follows:

$$\begin{aligned} N_D^+ &= \frac{N_D}{1 + g_D \exp\left[\left(\frac{E_F - E_d + q\phi}{k_B T}\right)\right]} \\ \Rightarrow \int_0^\phi N_D^+ d\phi' &= N_D \int_0^\phi \frac{1}{1 + g_D \exp\left[\left(\frac{E_F - E_d + q\phi'}{k_B T}\right)\right]} d\phi', \\ \text{put } x &= \left(\frac{E_F - E_d + q\phi'}{k_B T}\right) \text{ and use: } \int \frac{1}{1 + ae^x} dx = x - \ln(1 + ae^x) + C \\ \Rightarrow \int_0^\phi N_D^+ d\phi' &= N_D \frac{k_B T}{q} \left\{ \left(\frac{q\phi}{k_B T}\right) + \ln\left[\frac{1 + g_D \exp\left(\frac{E_F - E_d}{k_B T}\right)}{1 + g_D \exp\left(\frac{E_F - E_d + q\phi}{k_B T}\right)}\right] \right\} --- (1.13) \end{aligned}$$

$$\begin{aligned} \text{Similarly, } N_A^- &= \frac{N_A}{1 + g_A \exp\left[\left(\frac{E_a - E_F - q\phi}{k_B T}\right)\right]} \\ \Rightarrow \int_0^\phi N_A^- d\phi' &= N_A \frac{k_B T}{q} \left\{ \left(\frac{q\phi}{k_B T}\right) + \ln\left[\frac{1 + g_A \exp\left(\frac{E_a - E_F - q\phi}{k_B T}\right)}{1 + g_A \exp\left(\frac{E_a - E_F}{k_B T}\right)}\right] \right\} --- (1.14) \end{aligned}$$

$\int_0^\phi \rho(\phi') d\phi'$ can be calculated from equations 1.9, 1.10, 1.13 and 1.14 and we can get $E(\phi)$ relation for degenerate case.

$$E = -\frac{d\phi}{dx} = \text{sign}(\phi) \left\{ \frac{2k_B T}{\epsilon_0 K_{Si}} \left\{ N_V \left[\mathcal{F}_{3/2} \left(\frac{E_V - E_F - q\phi}{k_B T} \right) - \mathcal{F}_{3/2} \left(\frac{E_V - E_F}{k_B T} \right) \right] + N_C \left[\mathcal{F}_{3/2} \left(\frac{E_F - E_C + q\phi}{k_B T} \right) - \mathcal{F}_{3/2} \left(\frac{E_F - E_C}{k_B T} \right) \right] \right. \right. \\ \left. \left. - N_D \ln \left[\frac{1 + g_D \exp \left(\frac{E_F - E_d}{k_B T} \right)}{1 + g_D \exp \left(\frac{E_F - E_d + q\phi}{k_B T} \right)} \right] + N_A \ln \left[\frac{1 + g_A \exp \left(\frac{E_a - E_F - q\phi}{k_B T} \right)}{1 + g_A \exp \left(\frac{E_a - E_F}{k_B T} \right)} \right] - q(N_D - N_A) \frac{q\phi}{k_B T} \right\}^{1/2} \right\} \quad (1.15)$$

Above equation holds for semiconductor on right side. For semiconductor on left side, multiply the RHS of above equation by -1. (Exercise: Find out the expression for Debye length in the degenerate case.)

2. Equilibrium band diagram of pn junction:

The equilibrium band diagram is obtained by solving Poisson equation. This can be done by two methods which are described below.

A. Direct method to solve Poisson equation

Consider now a pn junction in equilibrium. The potential inside pn junction is determined by the Poisson equation. If we know the potential at the interface between p and n semiconductors (ϕ_s), we can solve differential equation 1.15 (degenerate case) or 1.8 (non-degenerate case) numerically to get potential as a function of x. The potential at the interface can be obtained from the boundary condition that electric displacement field ($D = \epsilon_0 K_S E$) is continuous i.e.

$$K_1 E(0^-) = K_2 E(0^+) \quad (2.1)$$

K_1 and K_2 are the dielectric constants of the p and n sides respectively. In the non-degenerate case, using equation 1.8, the interface potential (ϕ_s) can be obtained from the following equation:

$$K_1 \left\{ p_{p0} \left[\exp \left(-\frac{q\phi_s}{k_B T} \right) - 1 \right] + n_{p0} \left[\exp \left(\frac{q\phi_s}{k_B T} \right) - 1 \right] + N_A \frac{q\phi_s}{k_B T} \right\} \\ = K_2 \left\{ p_{n0} \left[\exp \left(-\frac{q(\phi_s - V_{bi})}{k_B T} \right) - 1 \right] + n_{n0} \left[\exp \left(\frac{q(\phi_s - V_{bi})}{k_B T} \right) - 1 \right] - N_D \frac{q(\phi_s - V_{bi})}{k_B T} \right\} \quad (2.2)$$

p_{n0} denotes the concentration of holes in bulk in the n-type semiconductor (right-side) etc. **Note that on the RHS, the ϕ_s term has been replaced by $\phi_s - V_{bi}$.** This is because we have taken the potential to be 0 on the LHS and V_{bi} on the right RHS. The equation 1.8 applies when the potential (band bending) is 0 in the bulk of semiconductor. Thus, while using equation 1.8 for the RHS semiconductor, we should take potential at $x=0$ (surface potential) as $\phi_s - V_{bi}$ and potential in the bulk as 0. (see figure below).

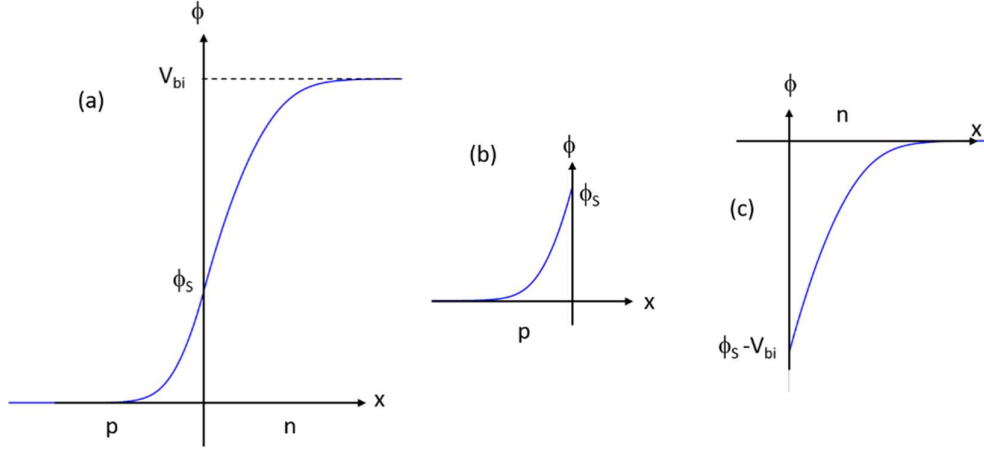


Figure 2. 1(a) Typical potential profile in a pn junction. (b) and (c) potentials in the LHS and RHS semiconductors respectively. Potential goes to 0 in the bulk of semiconductor.

The solution to above equation can be obtained by using fzero function in MATLAB. For homo-junctions (i.e. same semiconductor on both sides, $K_1=K_2$), the above equation can be solved to get an analytical expression for ϕ_s (i.e. we don't need to use fzero function of MATLAB) as follows:

put $p_{p0} = p_{n0} \exp(\frac{qV_{bi}}{k_B T})$ and $n_{p0} = n_{n0} \exp(\frac{-qV_{bi}}{k_B T})$ in eqn 2.2

we get, $\{-p_{p0} - n_{p0} + N_A \frac{q\phi_s}{k_B T}\} = \{-p_{n0} - n_{n0} - N_D \frac{q(\phi_s - V_{bi})}{k_B T}\}$

$$\Rightarrow \phi_s = \frac{N_D}{(N_A + N_D)} V_{bi} + \frac{1}{N_A + N_D} \frac{k_B T}{q} [p_{p0} + n_{p0} - (p_{n0} + n_{n0})] \quad \text{--- (2.3a)}$$

Now use, $n = \frac{N_D - N_A}{2} + \left[\left(\frac{N_D - N_A}{2} \right)^2 + n_i^2 \right]^{1/2}$ and $p = \frac{N_A - N_D}{2} + \left[\left(\frac{N_A - N_D}{2} \right)^2 + n_i^2 \right]^{1/2}$

we get, $\phi_s = \frac{N_D}{(N_A + N_D)} V_{bi} - \frac{k_B T}{q} \frac{[N_D^2 + 4n_i^2]^{1/2} - [N_A^2 + 4n_i^2]^{1/2}}{N_A + N_D} \quad \text{--- (2.3b)}$

The first term in the above equation is the interface potential that we get from the depletion approximation, and second term is correction over that. Proving equation 2.3b is an exercise problem in Ashcroft and Mermin!

Equations 2.2 and 2.3 for finding out interface potential hold for non-degenerate case. For degenerate case, we follow the same procedure i.e. use equation 2.1, but with electric field given by equation 1.15.

MATLAB program pn_junction_equi_solution_non_iterative.m is based on above algorithm. The solution is obtained for both degenerate case and non-degenerate approximation. The program needs band gap, effective density of states, donor and acceptor energy levels and their degeneracies, dielectric constant etc. for both sides. This is given in a separate input file. The input file calculates intrinsic carrier concentration (n_i), fermi level positions, built-in potential V_{bi} (difference between the Fermi levels) for both non-degenerate and degenerate cases. In the

non-degenerate case E_F (Fermi-level) is obtained from the equation block 1.2, whereas in the degenerate case E_F is obtained from equation block 1.1 using the condition that charge density must be 0 at the correct value of E_F . In the program we define a function to get charge density using equation block 1 and then use fzero function get value of E_F .

The program defines electric field (E) as a function of ϕ for both non-degenerate (equation 1.5) and degenerate cases (equation 1.15) and uses equation 2.1 to find out ϕ_s using fzero function of MATLAB. The program then solves the differential equation, $E = -d\phi/dx$ to get $\phi(x)$ using ode45 routine of MATLAB. Charge density, electric field, band diagram etc. are then obtained from $\phi(x)$. e.g. conduction band as a function of x on the LHS is given by $E_{c,L} - q\phi(x)$ and on the RHS is given by $E_{c,R} - q\phi(x)$. In the code, vacuum level before the two semiconductors are connected is taken at 0, and thus $E_{c,L} = -\chi_L$ and $E_{c,R} = -\chi_R$, where χ denotes the electron affinity. Various quantities are then plotted by calling a separate program. You can comment out the plots that are not required.

Some results obtained from above program are shown below. Figure 2.2 shows the potential and charge density inside Si pn junction at 300 K. The simulation results are compared with results from depletion approximation.

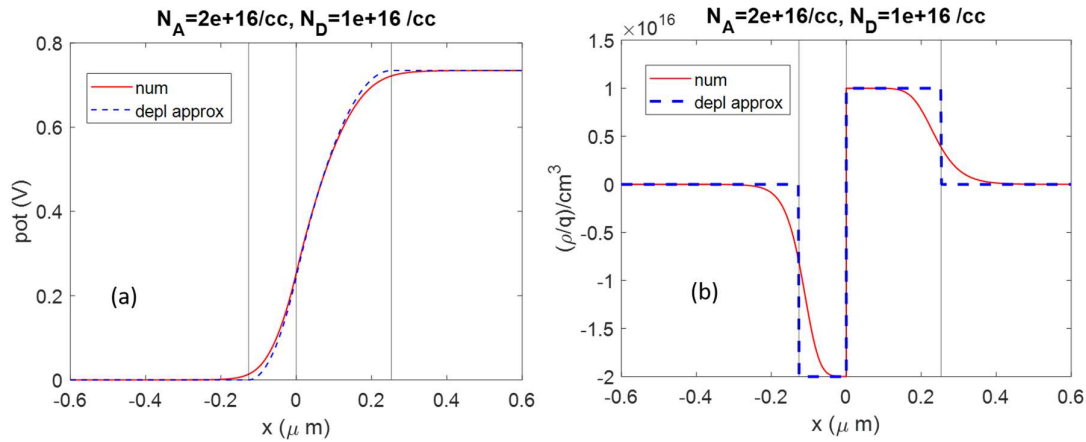


Figure 2.2 (a) potential and (b) charge density inside Si pn junction

Figure 2.3 (a) shows the charge density inside a Si p^+n junction. The narrow peak in the charge density near the interface on n-side is due to the inversion. Fig 2.3 b shows the band diagram of AlGaAs/GaAs junction (HEMT). These results are obtained using degenerate solution, as the Fermi-level lies within the conduction band near the interface on GaAs side. Note that on the GaAs side due to low doping, the Debye length is large and the band bending continues beyond the x-axis limit of $1 \mu\text{m}$.

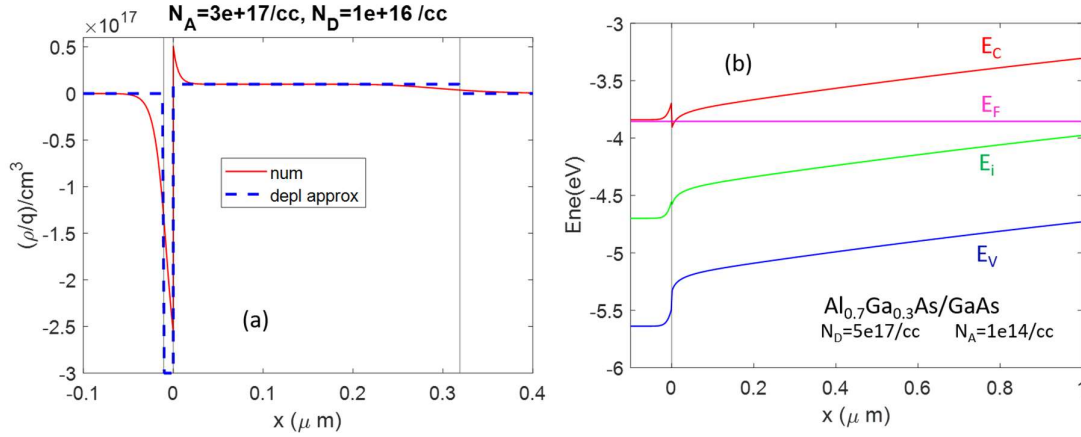


Figure 2.3 (a) charge density inside Si p^+n junction (b) energy band diagram of AlGaAs/GaAs junction

B. Iterative method to solve Poisson equation

The non-linear Poisson equation 1.4 can be solved numerically by using iterative method as well. Here we need to worry about the convergence, but the advantage is that this method can be easily used even for non-uniform doping. Another point is that the equation 1.5 is derived assuming semi-infinite semiconductor so that the electric field is zero at the upper limit of integral. If the length is finite (length < 5 Debye length), equation 1.5 cannot be used and iterative method would be convenient.

The Poisson equation 1.4 is applicable when the dielectric constant K ($\epsilon = \epsilon_0 K$) is independent of x . If ϵ depends on x , Poisson equation is written as:

$$\frac{dD}{dx} = \rho$$

$$\Rightarrow \frac{d}{dx} \left(\epsilon \frac{d\phi}{dx} \right) = -\rho \quad \text{--- (2.4)}$$

The first step in the iterative algorithm is to discretize the Poisson equation. The discretization length (Δx) in figure 2.3 should be chosen to be less than Debye length. Let's consider a pn junction with electrical permittivity of ϵ_1 and ϵ_2 on two sides. To understand the method, let's discretize this into 5 equally spaced points with point no. 3 at the interface. To solve equation 2.4, we need two boundary conditions: let's assume that we know potential at the two ends (ϕ_L and ϕ_R). This is shown schematically in the figure below:

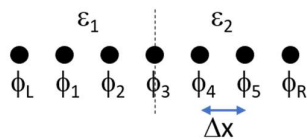


Figure 2.3 Discretization of potential inside a pn junction.

Note that the potential is defined at each discretized point (node), and the derivative of the potential is defined at the midpoints. The discretization steps and the final matrix equation are given below:

calculate $\varepsilon \frac{d\phi}{dx}$ at mid points

e.g. at 2.5, $\varepsilon \frac{d\phi}{dx} = \varepsilon_1 \frac{\phi_3 - \phi_2}{\Delta x}$ and at 3.5, $\varepsilon \frac{d\phi}{dx} = \varepsilon_2 \frac{\phi_4 - \phi_3}{\Delta x}$

$\rightarrow \frac{d}{dx} \varepsilon \frac{d\phi}{dx}$ at point 3 is $\frac{1}{\Delta x} [\varepsilon_2 \frac{\phi_4 - \phi_3}{\Delta x} - \varepsilon_1 \frac{\phi_3 - \phi_2}{\Delta x}] = \frac{1}{(\Delta x)^2} [\varepsilon_2 \phi_4 - (\varepsilon_1 + \varepsilon_2) \phi_3 + \varepsilon_1 \phi_2]$

We get following 5 eqns :

$$\left. \begin{aligned} f_1 &\equiv \varepsilon_1 \frac{\phi_L - 2\phi_1 + \phi_2}{(\Delta x)^2} + \rho(\phi_1) = 0 \quad \text{eqn at point 1} \\ f_2 &\equiv \varepsilon_1 \frac{\phi_1 - 2\phi_2 + \phi_3}{(\Delta x)^2} + \rho(\phi_2) = 0 \quad \text{eqn at point 2} \\ f_3 &\equiv \frac{1}{(\Delta x)^2} [\varepsilon_2 \phi_4 - (\varepsilon_1 + \varepsilon_2) \phi_3 + \varepsilon_1 \phi_2] + \rho(\phi_3) = 0 \quad \text{at point 3} \\ f_4 &\equiv \varepsilon_2 \frac{\phi_3 - 2\phi_4 + \phi_5}{(\Delta x)^2} + \rho(\phi_4) = 0 \quad \text{eqn at point 4} \\ f_5 &\equiv \varepsilon_2 \frac{\phi_4 - 2\phi_5 + \phi_R}{(\Delta x)^2} + \rho(\phi_5) = 0 \quad \text{eqn at point 5} \end{aligned} \right\} \text{--- (2.5a)}$$

define, $t_1 = \frac{\varepsilon_1}{(\Delta x)^2}$ and $t_2 = \frac{\varepsilon_2}{(\Delta x)^2}$ --- (2.6)

$$\Rightarrow \underbrace{\begin{pmatrix} -2t_1 & t_1 & 0 & 0 & 0 \\ t_1 & -2t_1 & t_1 & 0 & 0 \\ 0 & t_1 & -t_1 - t_2 & t_2 & 0 \\ 0 & 0 & t_2 & -2t_2 & t_2 \\ 0 & 0 & 0 & t_2 & -2t_2 \end{pmatrix}}_{\mathcal{J}_1} \begin{pmatrix} \phi_1 \\ \phi_2 \\ \phi_3 \\ \phi_4 \\ \phi_5 \end{pmatrix} = - \begin{pmatrix} \rho(\phi_1) \\ \rho(\phi_2) \\ \rho(\phi_3) \\ \rho(\phi_4) \\ \rho(\phi_5) \end{pmatrix} - \underbrace{\begin{pmatrix} t_1 \phi_L \\ 0 \\ 0 \\ 0 \\ t_2 \phi_R \end{pmatrix}}_{V_{bdy}} \text{--- (2.5b)}$$

As can be seen from above, values of ϕ_L and ϕ_R are required to write equations at the first and last point. (This where we use boundary conditions.) The terms arising from boundary conditions are denoted by V_{bdy} in equation 2.5b. ϕ_L and ϕ_R are taken as 0 and V_{bi} respectively, in the program. The set of 5 equations in equation block 2.5a can be written as: $f_i(\phi)=0$, with $i=1$ to 5 and $\phi=(\phi_1, \phi_2, \phi_3, \phi_4, \phi_5)$. Equations of this form can be solved by Newton-Raphson method. In this method, we need to start with an initial guess of ϕ , and iteratively obtain the solution. If the solution at step k is ϕ^k , the solution in the next step ($k+1$) is given by, $J_{ij}^k \Delta^k_j = -f_j(\phi^k)$, where $\Delta^k = \phi^{k+1} - \phi^k$ and the Jacobian is defined as, $J_{ij} = \partial f_i / \partial \phi_j$. For the set of 5 equations (equation block 2.5a) this procedure can be written as following:

$$\text{define, } \mathcal{J} = \begin{pmatrix} \left(\frac{\partial f_1}{\partial \phi_1}\right)_k & \left(\frac{\partial f_1}{\partial \phi_2}\right)_k & \cdot & \cdot & \cdot \\ \left(\frac{\partial f_2}{\partial \phi_1}\right)_k & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \left(\frac{\partial f_5}{\partial \phi_5}\right)_k \end{pmatrix} = \underbrace{\begin{pmatrix} -2t_1 & t_1 & 0 & 0 & 0 \\ t_1 & -2t_1 & t_1 & 0 & 0 \\ 0 & t_1 & -(t_1+t_2) & t_2 & 0 \\ 0 & 0 & t_2 & -2t_2 & t_2 \\ 0 & 0 & 0 & t_2 & -2t_2 \end{pmatrix}}_{\mathcal{J}_1 \text{ --- (2.7a)}} + \underbrace{\begin{pmatrix} \left.\frac{d\rho}{d\phi}\right|_1 & 0 & 0 & 0 & 0 \\ 0 & \left.\frac{d\rho}{d\phi}\right|_2 & 0 & 0 & 0 \\ 0 & 0 & \left.\frac{d\rho}{d\phi}\right|_3 & 0 & 0 \\ 0 & 0 & 0 & \left.\frac{d\rho}{d\phi}\right|_4 & 0 \\ 0 & 0 & 0 & 0 & \left.\frac{d\rho}{d\phi}\right|_5 \end{pmatrix}}_{\mathcal{J}_2 \text{ --- (2.7b)}}$$

$$[\mathcal{J}] \begin{pmatrix} \Delta_1 \\ \Delta_2 \\ \cdot \\ \cdot \\ \Delta_5 \end{pmatrix} = - \begin{pmatrix} f_1(\phi^k) \\ f_2(\phi^k) \\ \cdot \\ \cdot \\ f_5(\phi^k) \end{pmatrix} \quad (\text{Newton - Raphson method})$$

$$\rightarrow [\mathcal{J}_1 + \mathcal{J}_2] \left\{ \begin{pmatrix} \phi_1^{k+1} \\ \phi_2^{k+1} \\ \phi_3^{k+1} \\ \phi_4^{k+1} \\ \phi_5^{k+1} \end{pmatrix} - \begin{pmatrix} \phi_1^k \\ \phi_2^k \\ \phi_3^k \\ \phi_4^k \\ \phi_5^k \end{pmatrix} \right\} = - \left\{ \begin{pmatrix} -2t_1 & t_1 & 0 & 0 & 0 \\ t_1 & -2t_1 & t_1 & 0 & 0 \\ 0 & t_1 & -t_1-t_2 & t_2 & 0 \\ 0 & 0 & t_2 & -2t_2 & t_2 \\ 0 & 0 & 0 & t_2 & -2t_2 \end{pmatrix} \begin{pmatrix} \phi_1^k \\ \phi_2^k \\ \phi_3^k \\ \phi_4^k \\ \phi_5^k \end{pmatrix} + \begin{pmatrix} t_1 \phi_L \\ 0 \\ 0 \\ 0 \\ t_2 \phi_R \end{pmatrix} + \begin{pmatrix} \rho(\phi_1^k) \\ \rho(\phi_2^k) \\ \rho(\phi_3^k) \\ \rho(\phi_4^k) \\ \rho(\phi_5^k) \end{pmatrix} \right\}$$

$$\rightarrow [\mathcal{J}_1 + \mathcal{J}_2] \begin{pmatrix} \phi_1^{k+1} \\ \phi_2^{k+1} \\ \phi_3^{k+1} \\ \phi_4^{k+1} \\ \phi_5^{k+1} \end{pmatrix} = \mathcal{J}_2 \begin{pmatrix} \phi_1^k \\ \phi_2^k \\ \phi_3^k \\ \phi_4^k \\ \phi_5^k \end{pmatrix} - \underbrace{\begin{pmatrix} t_1 \phi_L \\ 0 \\ 0 \\ 0 \\ t_2 \phi_R \end{pmatrix}}_{V_{bdy}} - \begin{pmatrix} \rho(\phi_1^k) \\ \rho(\phi_2^k) \\ \rho(\phi_3^k) \\ \rho(\phi_4^k) \\ \rho(\phi_5^k) \end{pmatrix} \quad \text{--- (2.7c)}$$

--- (2.7d)

Thus, the solution at step k+1 is obtained from solution at step k from equation 2.7c. This can be done by multiplying inverse of J with the RHS of equation 2.7c. Efficient way is to use MATLAB's backslash operator to solve equation 2.7c. The iterations in Newton-Raphson method are carried out till the difference between ϕ at step k+1 and k is less than a certain value.

Note that we need to find out $d\rho/d\phi$ to get J_2 . For non-degenerate case this is given by:

$$\rho = q(p_0 \exp(-\frac{q\phi}{k_B T}) - n_0 \exp(\frac{q\phi}{k_B T}) + N_D - N_A)$$

$$\Rightarrow \frac{d\rho}{d\phi} = \frac{q^2}{k_B T} (-p_0 \exp(-\frac{q\phi}{k_B T}) - n_0 \exp(\frac{q\phi}{k_B T})) = \frac{-q^2}{k_B T} (p + n) \quad \text{--- (2.8)}$$

For the degenerate case, we need to use equation block 1.1 for ρ . The derivatives of the concentration of ionized donors and acceptors w.r.t ϕ are given by:

$$\left. \begin{aligned}
N_D^+ &= \frac{N_D}{1 + g_D \exp\left[\left(\frac{E_F - E_d + q\phi}{k_B T}\right)\right]}, \\
\Rightarrow \frac{dN_D^+}{d\phi} &= \frac{-N_D}{\{1 + g_D \exp\left[\left(\frac{E_F - E_d + q\phi}{k_B T}\right)\right]\}^2} g_D \exp\left[\left(\frac{E_F - E_d + q\phi}{k_B T}\right)\right] \frac{q}{k_B T} \\
&= \frac{-(N_D^+)^2}{N_D} \frac{N_D - N_D^+}{N_D^+} \frac{q}{k_B T} = \frac{-N_D^+}{N_D} (N_D - N_D^+) \frac{q}{k_B T} \\
N_A^- &= \frac{N_A}{1 + g_A \exp\left[\left(\frac{E_a - q\phi - E_F}{k_B T}\right)\right]} \Rightarrow \frac{dN_A^-}{d\phi} = \frac{N_A^-}{N_A} (N_A - N_A^-) \frac{q}{k_B T}
\end{aligned} \right\} \text{--- (2.9)}$$

The derivatives of n and p w.r.t ϕ are given by the following equations:

$$\left. \begin{aligned}
n &= N_c \mathcal{F}_{1/2}\left(\frac{E_F - E_C + q\phi}{k_B T}\right) \Rightarrow \frac{dn}{d\phi} = N_c \mathcal{F}_{-1/2}\left(\frac{E_F - E_C + q\phi}{k_B T}\right) * \frac{q}{k_B T} \\
p &= N_v \mathcal{F}_{1/2}\left(\frac{E_V - q\phi - E_F}{k_B T}\right) \Rightarrow \frac{dp}{d\phi} = N_v \mathcal{F}_{-1/2}\left(\frac{E_V + q\phi - E_F}{k_B T}\right) * \frac{-q}{k_B T}
\end{aligned} \right\} \text{--- (2.10)}$$

where $\mathcal{F}_{-1/2}$ is the Fermi-Dirac integral of order -1/2. We have used equation 1.11 for taking the derivative of $\mathcal{F}_{1/2}$.

Thus, we get in the degenerate case:

$$\left. \begin{aligned}
\frac{d\rho}{d\phi} &= q \frac{d}{d\phi} [p - n + N_D^+ - N_A^-] \\
&= \frac{-q^2}{k_B T} \left[N_v \mathcal{F}_{-1/2}\left(\frac{E_V + q\phi - E_F}{k_B T}\right) + N_c \mathcal{F}_{-1/2}\left(\frac{E_F - E_C + q\phi}{k_B T}\right) + \frac{N_D^+}{N_D} (N_D - N_D^+) + \frac{N_A^-}{N_A} (N_A - N_A^-) \right]
\end{aligned} \right\} \text{--- (2.11)}$$

Program `pn_junction_equi_solution_iterative.m` uses iterative algorithm to solve p-n junction equilibrium problem. The solution is obtained for both degenerate case and non-degenerate approximation. The program needs input file with information for two sides of pn junctions, same as the previous program. The results are plotted by calling a separate program.

In the code, you can put $J_2=0$ and use a mixing parameter β to obtain solution for the next step. Equation 2.7c with $J_2=0$ is the same as equation 2.5b, with the difference that in equation 2.7c the LHS is at step $k+1$ and RHS is at step k . You can try with small values of β to see if the solution converges.

Exercise problem: Modify the code to calculate the equilibrium configuration in a pin diode. You can do it using iterative as well as non-iterative method.

Exercise problem: Modify the code to include polarization charges at the interface.

Exercise problem: Simulate equilibrium band diagram of metal-semiconductor junction.

3. Calculation of dc current as a function of dc voltage in pn junction

We consider here pn homojunction (i.e. same semiconductor on both sides) under non-degenerate approximation. The calculation of current requires equations for electron and hole current densities, continuity equations and the Poisson equation. These equations are given below (See Pierret):

$$\left. \begin{aligned} J_n &= q \mu_n n E + q D_n \frac{dn}{dx} \\ J_p &= q \mu_p p E - q D_p \frac{dp}{dx} \\ J_C &= J_n + J_p \end{aligned} \right\} \text{--- (3.1a)}$$

$$\left. \begin{aligned} \frac{\partial n}{\partial t} &= \frac{1}{q} \frac{dJ_n}{dx} - R \\ \frac{\partial p}{\partial t} &= -\frac{1}{q} \frac{dJ_p}{dx} - R \end{aligned} \right\} \text{--- (3.1b)}$$

$$\text{where net recombination rate, } R = \frac{np - n_i^2}{\tau_p(n + n_i) + \tau_n(p + p_i)} \text{--- (3.1c)}$$

$$\frac{d}{dx} \epsilon \frac{d\phi}{dx} = -\rho \text{--- (3.1d),}$$

$$\text{where } \rho = q(p - n + N_D - N_A)$$

The symbols have their usual meaning. (Here we include only SRH recombination and generation.) In the previous case of pn junction in equilibrium, potential at each node was the only unknown quantity, which was determined by the Poisson equation. Here we have three unknowns: n, p and ϕ at each node, which can be determined by the two continuity equations (3.1b) and Poisson equation (3.1d). For numerical solution we need to discretize these three equations. Discretization of the Poisson equation has been done in the previous section. We will now to see how to discretize the continuity equations.

To discretize the continuity equation, we first need to discretize the current density equations (3.1a). Note that n, p and ϕ are defined at each node, whereas electric field and the current density are defined at the mid-points, as they involve derivatives of n, p and ϕ . A simple way to discretize electron and hole current densities is to write:

$$\left. \begin{aligned} J_{n,i+1/2} &= q \mu_n \frac{n_i + n_{i+1}}{2} \left(\frac{\phi_i - \phi_{i+1}}{\Delta x} \right) + q D_n \frac{n_{i+1} - n_i}{\Delta x} \\ J_{p,i+1/2} &= q \mu_p \frac{p_i + p_{i+1}}{2} \left(\frac{p_i - p_{i+1}}{\Delta x} \right) - q D_p \frac{p_{i+1} - p_i}{\Delta x} \end{aligned} \right\} \text{--- (3.2)}$$

This then can be inserted into equation 3.1b to discretize the two continuity equations. The above scheme however needs Δx to be quite small. To overcome this problem, we will use the Scharfetter-Gummel discretization Scheme. In this scheme, the J_n and J_p are discretized as follows (see appendix for derivation):

$$\left. \begin{aligned} J_{n,i+1/2} &= q \frac{D_{n,i+1/2}}{\Delta x} \{n_{i+1} B(z) - n_i B(-z)\} \\ J_{p,i+1/2} &= -q \frac{D_{p,i+1/2}}{\Delta x} \{p_{i+1} B(-z) - p_i B(z)\} \\ \text{where Brillouin function, } B(z) &= \frac{z}{e^z - 1} \text{ and } z = \frac{\phi_{i+1} - \phi_i}{V_{ther}} \end{aligned} \right\} \text{--- (3.3)}$$

(Exercise: By Taylor series expansion of Brillouin function, $B(z) \cong 1 - z/2$, show that above equation reduces to equation 3.2.) Note that in equation 3.3 and 3.2, the index i runs from 0 to N , where N = number of discretized points. n_0 and p_0 denote the electron and hole concentrations at the left boundary (i.e. $n_0 = n_L$, $p_0 = p_L$), and n_{N+1} and p_{N+1} denote the electron and hole concentrations at the right boundary (i.e. $n_{N+1} = n_R$, $p_{N+1} = p_R$). The current densities are defined midway between the points.

Further writing,

$$\begin{aligned} \left(\frac{dJ_n}{dx} \right)_i &= \frac{J_{n,i+1/2} - J_{n,i-1/2}}{\Delta x} \text{--- (3.4a)} \\ &= \frac{q}{(\Delta x)^2} \{ D_{n,i+1/2} [n_{i+1} B(\frac{\phi_{i+1} - \phi_i}{V_{ther}}) - n_i B(-\frac{\phi_{i+1} - \phi_i}{V_{ther}})] - D_{n,i-1/2} [n_i B(\frac{\phi_i - \phi_{i-1}}{V_{ther}}) - n_{i-1} B(-\frac{\phi_i - \phi_{i-1}}{V_{ther}})] \} \\ \text{and } \left(\frac{dJ_p}{dx} \right)_i &= \frac{J_{p,i+1/2} - J_{p,i-1/2}}{\Delta x} \text{--- (3.4b)} \\ &= \frac{q}{(\Delta x)^2} \{ -D_{p,i+1/2} [p_{i+1} B(-\frac{\phi_{i+1} - \phi_i}{V_{ther}}) - p_i B(\frac{\phi_{i+1} - \phi_i}{V_{ther}})] + D_{p,i-1/2} [p_i B(-\frac{\phi_i - \phi_{i-1}}{V_{ther}}) - p_{i-1} B(\frac{\phi_i - \phi_{i-1}}{V_{ther}})] \} \end{aligned}$$

we get discretized continuity equations as:

$$\begin{aligned} \frac{1}{q} \frac{dJ_n}{dx} - R &= \frac{\partial n}{\partial t} \text{ becomes :} \\ \frac{1}{q} \frac{J_{n,i+1/2} - J_{n,i-1/2}}{\Delta x} - R_i &= \frac{\partial n_i}{\partial t} \\ - > \frac{1}{(\Delta x)^2} \{ n_{i+1} [D_{n,i+1/2} B(\frac{\phi_{i+1} - \phi_i}{V_T})] - n_i [D_{n,i+1/2} B(\frac{\phi_i - \phi_{i+1}}{V_T}) + D_{n,i-1/2} B(\frac{\phi_i - \phi_{i-1}}{V_T})] + n_{i-1} [D_{n,i-1/2} B(\frac{\phi_{i-1} - \phi_i}{V_T})] \} - R(n_i, p_i) &= \frac{\partial n_i}{\partial t} \end{aligned} \left\} \text{--- (3.5a)}$$

$$\begin{aligned} -\frac{1}{q} \frac{dJ_p}{dx} - R &= \frac{\partial p}{\partial t} \text{ becomes :} \\ -\frac{1}{q} \frac{J_{p,i+1/2} - J_{p,i-1/2}}{\Delta x} - R_i &= \frac{\partial p_i}{\partial t} \\ - > \frac{1}{(\Delta x)^2} \{ p_{i+1} [D_{p,i+1/2} B(\frac{\phi_i - \phi_{i+1}}{V_T})] - p_i [D_{p,i+1/2} B(\frac{\phi_{i+1} - \phi_i}{V_T}) + D_{p,i-1/2} B(\frac{\phi_i - \phi_{i-1}}{V_T})] + p_{i-1} [D_{p,i-1/2} B(\frac{\phi_{i-1} - \phi_i}{V_T})] \} - R(n_i, p_i) &= \frac{\partial p_i}{\partial t} \end{aligned} \left\} \text{--- (3.5b)}$$

The index i runs from 1 to N in equations 3.4 and 3.5. The discretized Poisson equation (with the same dielectric constant on p and n sides) can be written as:

$$\frac{\epsilon}{(\Delta x)^2} [\phi_{i-1} - 2\phi_i + \phi_{i+1}] + \rho_i = 0 \text{--- (3.6)}$$

where $\rho_i = q(p_i - n_i + N_D - N_A)$

In equation 3.6, index i runs for 1 to N , and $\phi_0=\phi_L$ and $\phi_{N+1}=\phi_R$. If we discretize pn junction into N points, we have $3N$ unknowns (n , p and ϕ at each point) and we need $3N$ equations to solve for them. These $3N$ equations are: Poisson (equation 3.6), electron continuity (equation 3.5a) and hole continuity (equation 3.5b) equations. We are now going to scale the variables (n , p and ϕ) so that their variation is in the same range. This helps in solving the equations numerically. Let's define,

$$\tilde{n} = \frac{n}{N_1}, \tilde{p} = \frac{p}{N_2} \text{ and } \tilde{\phi} = \frac{\phi}{\phi_N} \quad \text{--- (3.7a)}$$

$$\tilde{D}_n = \frac{D_n}{D_{n0}}, \tilde{D}_p = \frac{D_p}{D_{p0}} \quad \text{--- (3.7b)}$$

$$U = \frac{\phi}{V_{therm}} \quad \text{--- (3.7c)}$$

where N_1 , N_2 and ϕ_N are normalization factors. You can choose them to be N_D , N_A , V_{therm} respectively in the code or any other values. We have also defined normalized the diffusion coefficients (equation 3.7b) by dividing them by value of diffusion coefficient at 0 electric field. Further we are going to multiply both sides of equations 3.5a and 3.5b by τ_1 and τ_2 respectively. These are scaling factors and you can choose to be τ_n and τ_p in the code or any other values. We have also defined dimensionless parameter U as ratio of potential to thermal voltage. U and $\tilde{\phi}$ would be same if we choose $\phi_N=V_{therm}$.

With these definitions, we can write discretized continuity equations 3.5 as:

$$\frac{D_{n0}\tau_1}{(\Delta x)^2} \{ \tilde{n}_{i+1} [\tilde{D}_{n,i+1/2} B(U_{i+1} - U_i)] - \tilde{n}_i [\tilde{D}_{n,i+1/2} B(U_i - U_{i+1}) + \tilde{D}_{n,i-1/2} B(U_i - U_{i-1})] + \tilde{n}_{i-1} [\tilde{D}_{n,i-1/2} B(U_{i-1} - U_i)] \} - \frac{\tau_1}{N_1} R(\tilde{n}_i, \tilde{p}_i) = \tau_1 \frac{\partial \tilde{n}_i}{\partial t} \quad \text{--- (3.8a)}$$

$$\frac{D_{p0}\tau_2}{(\Delta x)^2} \{ \tilde{p}_{i+1} [\tilde{D}_{p,i+1/2} B(U_i - U_{i+1})] - \tilde{p}_i [\tilde{D}_{p,i+1/2} B(U_{i+1} - U_i) + \tilde{D}_{p,i-1/2} B(U_i - U_{i-1})] + \tilde{p}_{i-1} [\tilde{D}_{p,i-1/2} B(U_{i-1} - U_i)] \} - \frac{\tau_2}{N_2} R(\tilde{n}_i, \tilde{p}_i) = \tau_2 \frac{\partial \tilde{p}_i}{\partial t} \quad \text{--- (3.8b)}$$

By defining,

$$\left. \begin{aligned} t_n &= \frac{L_n^2}{(\Delta x)^2}, \quad L_n^2 = D_{n0}\tau_1 \text{ and } R_n = \frac{\tau_1}{N_1} R \\ t_p &= \frac{L_p^2}{(\Delta x)^2}, \quad L_p^2 = D_{p0}\tau_2 \text{ and } R_p = \frac{\tau_2}{N_2} R \end{aligned} \right\} \quad \text{--- (3.9)}$$

equations 3.8 can be written in a simplified form as:

$$t_n \{ \tilde{n}_{i+1} [\tilde{D}_{n,i+1/2} B(U_{i+1} - U_i)] - \tilde{n}_i [\tilde{D}_{n,i+1/2} B(U_i - U_{i+1}) + \tilde{D}_{n,i-1/2} B(U_i - U_{i-1})] + \tilde{n}_{i-1} [\tilde{D}_{n,i-1/2} B(U_{i-1} - U_i)] \} - R_n(\tilde{n}_i, \tilde{p}_i) = \tau_1 \frac{\partial \tilde{n}_i}{\partial t} \quad \text{--- (3.10a)}$$

$$t_p \{ \tilde{p}_{i+1} [\tilde{D}_{p,i+1/2} B(U_i - U_{i+1})] - \tilde{p}_i [\tilde{D}_{p,i+1/2} B(U_{i+1} - U_i) + \tilde{D}_{p,i-1/2} B(U_i - U_{i-1})] + \tilde{p}_{i-1} [\tilde{D}_{p,i-1/2} B(U_{i-1} - U_i)] \} - R_p(\tilde{n}_i, \tilde{p}_i) = \tau_2 \frac{\partial \tilde{p}_i}{\partial t} \quad \text{--- (3.10b)}$$

The Poisson equation 3.6 is scaled as follows. We scale the doping profile, N_D-N_A by N , whose value can be chosen in the code. The scaled Poisson equation is obtained as:

$$eqn\ 3.6 \rightarrow \frac{\varepsilon}{(\Delta x)^2} \phi_N (\tilde{\phi}_{i-1} - 2\tilde{\phi}_i + \tilde{\phi}_{i+1}) + qN(\tilde{p}_i \frac{N_2}{N} - \tilde{n}_i \frac{N_1}{N} + \frac{N_D - N_A}{N}) = 0$$

$$define, t_{poi} = \frac{L_D^2}{(\Delta x)^2}, L_D = \sqrt{\frac{\varepsilon \phi_N}{qN}} \text{ and } \tilde{\rho}_i = (\tilde{p}_i \frac{N_2}{N} - \tilde{n}_i \frac{N_1}{N} + \frac{N_D - N_A}{N}) \text{ --- (3.11 a)}$$

$$\rightarrow t_{poi} (\tilde{\phi}_{i-1} - 2\tilde{\phi}_i + \tilde{\phi}_{i+1}) + \tilde{\rho}_i = 0 \text{ --- (3.11 b)}$$

To see how the equations 3.10 and 3.11 work, let's discretize semiconductor into 3 points. We also need 6 boundary conditions: (n, p and ϕ) at the two end points. This is shown schematically below. The values of n, p and ϕ at boundary points have also been scaled (equation 3.7a).

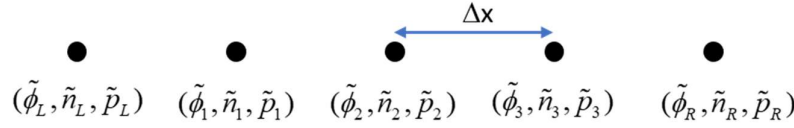


Figure 3.1 discretization of potential, electron and hole concentration inside semiconductor

The 9 equations that we need to solve are given below:

$$\left. \begin{aligned} f_1 &\equiv t_n \{ \tilde{n}_2 [\tilde{D}_{n,1.5} B(U_2 - U_1)] - \tilde{n}_1 [\tilde{D}_{n,1.5} B(U_1 - U_2) + \tilde{D}_{n,0.5} B(U_1 - U_L)] + \tilde{n}_L [\tilde{D}_{n,0.5} B(U_L - U_1)] \} - R_n(\tilde{n}_1, \tilde{p}_1) = \tau_1 \frac{\partial \tilde{n}_1}{\partial t} \\ f_2 &\equiv t_n \{ \tilde{n}_3 [\tilde{D}_{n,2.5} B(U_3 - U_2)] - \tilde{n}_2 [\tilde{D}_{n,2.5} B(U_2 - U_3) + \tilde{D}_{n,1.5} B(U_2 - U_1)] + \tilde{n}_1 [\tilde{D}_{n,1.5} B(U_1 - U_2)] \} - R_n(\tilde{n}_2, \tilde{p}_2) = \tau_1 \frac{\partial \tilde{n}_2}{\partial t} \\ f_3 &\equiv t_n \{ \tilde{n}_R [\tilde{D}_{n,3.5} B(U_R - U_3)] - \tilde{n}_3 [\tilde{D}_{n,3.5} B(U_3 - U_R) + \tilde{D}_{n,2.5} B(U_3 - U_2)] + \tilde{n}_2 [\tilde{D}_{n,2.5} B(U_2 - U_3)] \} - R_n(\tilde{n}_3, \tilde{p}_3) = \tau_1 \frac{\partial \tilde{n}_3}{\partial t} \end{aligned} \right\} 3.12a \text{ (electron)}$$

$$\left. \begin{aligned} f_4 &\equiv t_p \{ \tilde{p}_2 [\tilde{D}_{p,1.5} B(U_1 - U_2)] - \tilde{p}_1 [\tilde{D}_{p,1.5} B(U_2 - U_1) + \tilde{D}_{p,1/2} B(U_L - U_1)] + \tilde{p}_L [\tilde{D}_{p,1/2} B(U_1 - U_L)] \} - R_p(\tilde{n}_1, \tilde{p}_1) = \tau_2 \frac{\partial \tilde{p}_1}{\partial t} \\ f_5 &\equiv t_p \{ \tilde{p}_3 [\tilde{D}_{p,2.5} B(U_2 - U_3)] - \tilde{p}_2 [\tilde{D}_{p,2.5} B(U_3 - U_2) + \tilde{D}_{p,1.5} B(U_2 - U_1)] + \tilde{p}_1 [\tilde{D}_{p,1.5} B(U_1 - U_2)] \} - R_p(\tilde{n}_2, \tilde{p}_2) = \tau_2 \frac{\partial \tilde{p}_2}{\partial t} \\ f_6 &\equiv t_p \{ \tilde{p}_R [\tilde{D}_{p,3.5} B(U_3 - U_R)] - \tilde{p}_3 [\tilde{D}_{p,3.5} B(U_R - U_3) + \tilde{D}_{p,2.5} B(U_2 - U_3)] + \tilde{p}_2 [\tilde{D}_{p,2.5} B(U_3 - U_2)] \} - R_p(\tilde{n}_3, \tilde{p}_3) = \tau_2 \frac{\partial \tilde{p}_3}{\partial t} \end{aligned} \right\} 3.12b \text{ (hole)}$$

$$\left. \begin{aligned} f_7 &\equiv t_{poi} (\tilde{\phi}_L - 2\tilde{\phi}_1 + \tilde{\phi}_2) + \tilde{\rho}_1 = 0 \\ f_8 &\equiv t_{poi} (\tilde{\phi}_1 - 2\tilde{\phi}_2 + \tilde{\phi}_3) + \tilde{\rho}_2 = 0 \\ f_9 &\equiv t_{poi} (\tilde{\phi}_2 - 2\tilde{\phi}_3 + \tilde{\phi}_R) + \tilde{\rho}_3 = 0 \end{aligned} \right\} 3.12c \text{ (Poisson)}$$

We have divided these equations in three parts: electron block corresponds to discretized electron current density continuity equation, hole block corresponds to discretized hole current density continuity equation, and Poisson block corresponds to discretized Poisson equation. Note that to write the first and last equations in each block, we need to use the boundary conditions.

In matrix form these equations can be written as:

$$f \equiv \begin{pmatrix} -t_n[\tilde{D}_{n,1,5}\tilde{B}_{12} + \tilde{D}_{n,3/2}\tilde{B}_{1L}] & t_n\tilde{D}_{n,1,5}\tilde{B}_{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ t_n\tilde{D}_{n,1,5}\tilde{B}_{12} & -t_n[\tilde{D}_{n,2,5}\tilde{B}_{21} + \tilde{D}_{n,1,5}\tilde{B}_{21}] & t_n\tilde{D}_{n,2,5}\tilde{B}_{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & t_n\tilde{D}_{n,2,5}\tilde{B}_{21} & -t_n[\tilde{D}_{n,3,5}\tilde{B}_{1R} + \tilde{D}_{n,2,5}\tilde{B}_{12}] & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -t_p[\tilde{D}_{p,1,5}\tilde{B}_{21} + \tilde{D}_{p,3/2}\tilde{B}_{1L}] & t_p\tilde{D}_{p,1,5}\tilde{B}_{12} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & t_p\tilde{D}_{p,1,5}\tilde{B}_{21} & -t_p[\tilde{D}_{p,2,5}\tilde{B}_{12} + \tilde{D}_{p,1,5}\tilde{B}_{12}] & t_p\tilde{D}_{p,2,5}\tilde{B}_{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & t_p\tilde{D}_{p,2,5}\tilde{B}_{12} & -t_p[\tilde{D}_{p,3,5}\tilde{B}_{1R} + \tilde{D}_{p,2,5}\tilde{B}_{12}] & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -2t_{psol} & t_{psol} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & t_{psol} & -2t_{psol} & t_{psol} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & t_{psol} & -2t_{psol} & t_{psol} \end{pmatrix} \begin{pmatrix} \tilde{n}_1 \\ \tilde{n}_2 \\ \tilde{n}_3 \\ \tilde{p}_1 \\ \tilde{p}_2 \\ \tilde{p}_3 \\ \tilde{\phi}_1 \\ \tilde{\phi}_2 \\ \tilde{\phi}_3 \end{pmatrix} \quad \text{--- (3.13)}$$

where B_{ij} is defined as $B_{ij} = B(U_i - U_j)$. Note that these equations are coupled to each other. If you see the Poisson block, the charge density on right-hand side depends on electron and hole concentrations. If you see the electron block, f_1 , f_2 and f_3 depend on potential and the net recombination R depends on hole concentration. Same applies to the hole block. The above equation can be written in a compact form as:

$$f \equiv [\mathcal{J}_1] \begin{pmatrix} \tilde{n} \\ \tilde{p} \\ \tilde{\phi} \end{pmatrix} - \begin{pmatrix} R_n \\ R_p \\ -\tilde{\rho} \end{pmatrix} + \underbrace{\begin{pmatrix} n_{bdy} \\ p_{bdy} \\ V_{bdy} \end{pmatrix}}_{bdy} = \begin{pmatrix} \tau_1 \partial \tilde{n} / \partial t \\ \tau_2 \partial \tilde{p} / \partial t \\ 0 \end{pmatrix}, \quad \text{--- (3.14)}$$

$$\text{where } \mathcal{J}_1 = \begin{pmatrix} M_1 & 0 & 0 \\ 0 & M_2 & 0 \\ 0 & 0 & M_3 \end{pmatrix}, R_n = \begin{pmatrix} R_{n1} \\ R_{n2} \\ R_{n3} \end{pmatrix}, R_p = \begin{pmatrix} R_{p1} \\ R_{p2} \\ R_{p3} \end{pmatrix}, \tilde{\rho} = \begin{pmatrix} \tilde{\rho}_1 \\ \tilde{\rho}_2 \\ \tilde{\rho}_3 \end{pmatrix}$$

The matrices M_1 , M_2 and M_3 are marked in equation 3.13. The term bdy arises from the values of $\tilde{n}$, $\tilde{p}$ and $\tilde{\phi}$ at the boundaries. The term V_{bdy} is normalized form of equation 2.7d.

In steady state, we put $\partial/\partial t=0$, and thus the RHS is set to 0.

Boundary conditions: we need 6 boundary conditions as shown in figure 3. We assume ohmic contacts i.e. at the contacts the electrons and holes have the same Fermi-level and their concentration is same as in equilibrium. The potential can be taken as 0 on left end and $V_{bi}-V_A$ on right end, where V_A is the applied bias voltage. Thus, the boundary conditions on the scaled variables are:

$$\left. \begin{aligned} \tilde{n}_L &= n_{0L} / N_1, \tilde{p}_L = p_{0L} / N_1 \\ \tilde{n}_R &= n_{0R} / N_2, \tilde{p}_R = p_{0R} / N_2 \\ \tilde{\phi}_L &= 0, \tilde{\phi}_R = (V_{bi} - V_A) / \phi_N \end{aligned} \right\} \quad \text{--- (3.15)}$$

These 3N (N= no of discretization points) equations (equations 3.12 with $\partial/\partial t=0$) are of the form $f_i(\tilde{n}, \tilde{p}, \tilde{\phi}) = 0$, can be solved iteratively by using Newton-Raphson method. This requires at each step the 3N x 3N Jacobian matrix $J_{ij} = \partial f_i / \partial v_{j,}$, where $v=(n,p,\phi)$. This matrix is of the following form:

$$\mathcal{J} = \underbrace{\begin{pmatrix} M_1 & 0 & 0 \\ 0 & M_2 & 0 \\ 0 & 0 & M_3 \end{pmatrix}}_{\mathcal{J}_1} + \underbrace{\begin{pmatrix} -\frac{\partial R_n}{\partial \tilde{n}} 1 & -\frac{\partial R_n}{\partial \tilde{p}} 1 & \left[\frac{\partial f_n}{\partial \tilde{\phi}}\right] \\ -\frac{\partial R_p}{\partial \tilde{n}} 1 & -\frac{\partial R_p}{\partial \tilde{p}} 1 & \left[\frac{\partial f_p}{\partial \tilde{\phi}}\right] \\ -\frac{N_1}{N} 1 & \frac{N_2}{N} 1 & [0] \end{pmatrix}}_{\mathcal{J}_2} \quad \text{--- (3.16)}$$

As explained before, we can obtain solution at step k+1 from solution at step k using: $J_{ij}^k \Delta_j = -f_i^k$, where $\Delta = f^{k+1} - f^k$. The iterations are continued till convergence is achieved.

$$\begin{aligned} [\mathcal{J}_1^k + \mathcal{J}_2^k] \left\{ \begin{pmatrix} \tilde{n}^{k+1} \\ \tilde{p}^{k+1} \\ \tilde{\phi}^{k+1} \end{pmatrix} - \begin{pmatrix} \tilde{n}^k \\ \tilde{p}^k \\ \tilde{\phi}^k \end{pmatrix} \right\} &= -f^k = -\mathcal{J}_1^k \begin{pmatrix} \tilde{n}^k \\ \tilde{p}^k \\ \tilde{\phi}^k \end{pmatrix} + \begin{pmatrix} R_n^k \\ R_p^k \\ -\tilde{\rho}^k \end{pmatrix} - \begin{pmatrix} n_{bdy}^k \\ p_{bdy}^k \\ V_{bdy}^k \end{pmatrix} \\ \rightarrow [\mathcal{J}_1^k + \mathcal{J}_2^k] \begin{pmatrix} \tilde{n}^{k+1} \\ \tilde{p}^{k+1} \\ \tilde{\phi}^{k+1} \end{pmatrix} &= \mathcal{J}_2^k \begin{pmatrix} \tilde{n}^k \\ \tilde{p}^k \\ \tilde{\phi}^k \end{pmatrix} + \begin{pmatrix} R_n^k \\ R_p^k \\ -\tilde{\rho}^k \end{pmatrix} - \begin{pmatrix} n_{bdy}^k \\ p_{bdy}^k \\ V_{bdy}^k \end{pmatrix} \quad \text{--- (3.17)} \end{aligned}$$

The above method of solving is called as Newton's method. This involves solving 3N equations at each step, which can be computationally costly. Instead, we can use Gummel method where we solve for each block (electron, hole and Poisson) separately. If we know n, p and ϕ at step k, we first solve the following equations separately using Newton-Raphson method to get n and p at step k+1.

$$f_n \equiv [M_1](\tilde{n}) - R_n + n_{bdy} = 0 \rightarrow \mathcal{J}_e = M_1 - \frac{\partial R_n}{\partial \tilde{n}} 1 \quad \text{--- (3.18a)}$$

$$f_p \equiv [M_2](\tilde{p}) - R_p + p_{bdy} = 0 \rightarrow \mathcal{J}_h = M_2 - \frac{\partial R_p}{\partial \tilde{p}} 1 \quad \text{--- (3.18b)}$$

The values of n and p at step k+1 and ϕ at step k are then used to solve the Poisson block given by the following equation, using Newton-Raphson method:

$$\left. \begin{aligned} f_{poi} &\equiv [M_3](\tilde{\phi}) + \tilde{\rho} + V_{bdy} = 0 \\ \mathcal{J}_{poi} &= M_3 - \left(\frac{N_2}{N} \tilde{p} + \frac{N_1}{N} \tilde{n} \right) \frac{\phi_N}{V_{therm}} \end{aligned} \right\} \quad \text{--- (3.19)}$$

Note that the Jacobian for Poisson block is the same as was used before for non-degenerate case (see equations 2.7 and 2.8), but is in normalized form here. Thus, we get n, p and ϕ at step k+1 starting from step k. The process is continued till we convergence is achieved.

To re-clarify the use of Newton-Raphson method for solving equations 3.18 a, b and 3.19, below we write the steps to obtain the solution:

solution of eqn 3.18a (electron block):

$$(J_1^k + J_2^k)(\tilde{n}^{k+1} - \tilde{n}^k) = -f_n^k = -(J_1^k \tilde{n}^k - R_n^k + n_{bdy}^k), \text{ where } J_1^k = M_1^k \text{ and } J_2^k = -\frac{\partial R_n^k}{\partial \tilde{n}} \quad 1$$

$$-> (J_1^k + J_2^k) \tilde{n}^{k+1} = J_2^k \tilde{n}^k + R_n - n_{bdy} \quad \text{--- (3.20)}$$

Similar steps applied to hole (equation 3.18b) equation gives:

solution of eqn 3.18b (hole block):

$$(J_1^k + J_2^k)(\tilde{p}^{k+1} - \tilde{p}^k) = -f_p^k = -(J_1^k \tilde{p}^k - R_p^k + p_{bdy}^k), \text{ where } J_1^k = M_2^k \text{ and } J_2^k = -\frac{\partial R_p^k}{\partial \tilde{p}} \quad 1$$

$$-> (J_1^k + J_2^k) \tilde{p}^{k+1} = J_2^k \tilde{p}^k + R_p - p_{bdy} \quad \text{--- (3.21)}$$

The steps for Poisson eqn:

solution of eqn 3.19 (Poisson):

$$(J_1^k + J_2^k)(\tilde{\phi}^{k+1} - \tilde{\phi}^k) = -f_{poi}^k = -(J_1^k \tilde{\phi}^k + \tilde{\rho}^k + V_{bdy}^k), \text{ where } J_1^k = M_3^k \text{ and } J_2^k = -(\frac{N_2}{N} \tilde{p}^k + \frac{N_1}{N} \tilde{n}^k) \frac{\phi_N}{V_{therm}}$$

$$-> (J_1^k + J_2^k) \tilde{\phi}^{k+1} = J_2^k \tilde{\phi}^k - \tilde{\rho}^k - V_{bdy} \quad \text{--- (3.22)}$$

Once convergence is achieved either by Newton or Gummel method, current density is calculated using:

$$J_{n,i+1/2} = q \frac{D_{n,i+0.5}}{\Delta x} N_1 \{ \tilde{n}_{i+1} B(\frac{\tilde{\phi}_{i+1} - \tilde{\phi}_i}{V_{therm}} \phi_N) - \tilde{n}_i B(\frac{\tilde{\phi}_i - \tilde{\phi}_{i+1}}{V_{therm}} \phi_N) \} \quad \text{--- (3.23a)}$$

$$J_{p,i+1/2} = -q \frac{D_{p,i+0.5}}{\Delta x} N_2 \{ \tilde{p}_{i+1} B(\frac{\tilde{\phi}_i - \tilde{\phi}_{i+1}}{V_{therm}} \phi_N) - \tilde{p}_i B(\frac{\tilde{\phi}_{i+1} - \tilde{\phi}_i}{V_{therm}} \phi_N) \} \quad \text{--- (3.23b)}$$

Program pnjunction_IV_and_YV.m can be used to calculate current using either Newton or Gummel method. You can comment out Newton or Gummel method whichever method you do not want to use. It is important that the initial guess of n, p and ϕ is close to the actual values. The program first calculates n, p and ϕ in equilibrium. Then a small voltage is applied and program calculates n, p and ϕ using equilibrium values as starting point. Once convergence is achieved, current is calculated. The voltage is then again increased by a small step. Starting from the previously converged n, p and ϕ as initial values, program then calculates new n, p and ϕ , and then the resultant current at the new voltage point. The same program can be used to calculate admittance at each voltage point for a fixed frequency (see the next section). If you do not want this calculation, comment out the admittance calculation part.

Some results obtained for Si pn junctions at 300 K using this program are plotted below. The red circles in figure 3.1 show the current density vs voltage obtained numerically. The magenta circles show numerical result when the SRH RG process is turned off in the depletion region (RG process is turned off assuming that $\tau_n \rightarrow \infty$ and $\tau_p \rightarrow \infty$ inside the depletion region.) At high bias voltage red and magenta circles overlap. The blue line is obtained from the ideal diode equation and fits the magenta circles till about 0.65 V. The green line is obtained by adding RG

current contribution to ideal diode current (see equation 6.45 in Semiconductor Device Fundamentals by Pierret) and fits the red circles till about 0.65 V. We can see saturation in the current density at high bias voltages. As per diode theory, the log of current vs $qV_A/k_B T$ plot has slope of 1/2 at low bias voltage, which is followed by region of slope 1 (ideal diode), then again region with slope of 1/2 (high injection effect) and then saturation (see Pierret).

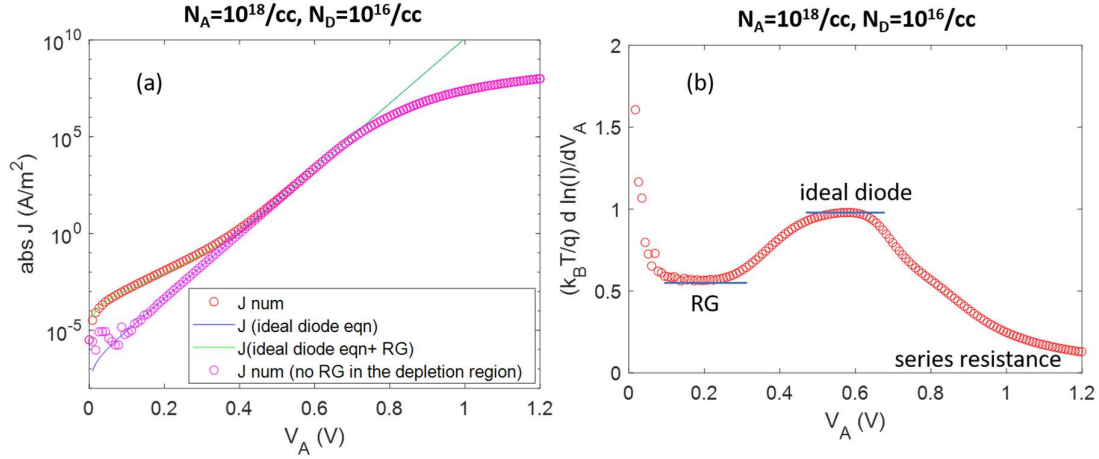


Figure 3.1(a) I-V characteristics of Si pn junction. Numerical results are compared with theory. (b) The derivative of log of current vs V_A . Different regions are indicated in the figure.

The derivative of log of current obtained numerically (red circles in figure 3.1.a) by V_A is plotted in fig 3.1.b. The region of 0.5 slope corresponds to the RG dominated current and region with slope 1 is the ideal diode behaviour dominated by diffusion current. At high voltages the slope becomes smaller due to the series resistance effect. The region with slope 0.5 due to the high injection effect is not seen in this plot.

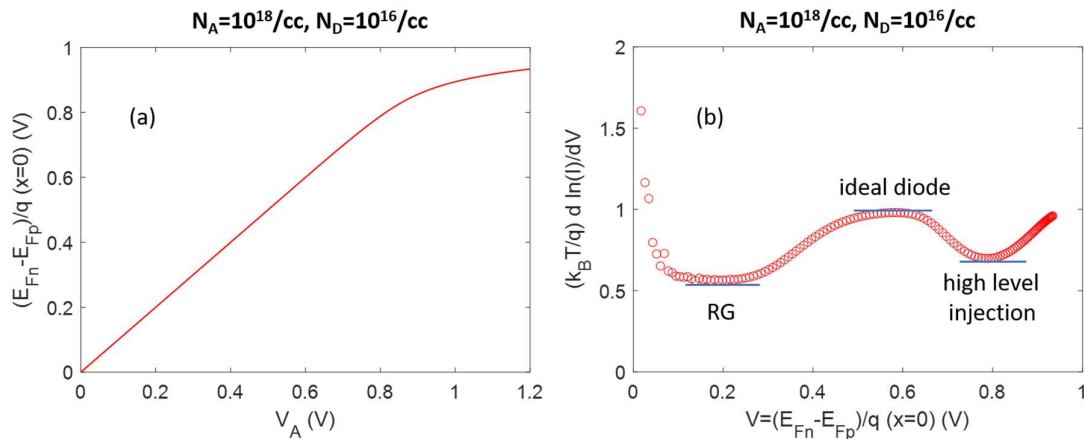


Figure 3.2 (a) difference in quasi-Fermi levels vs V_A . (b) same as figure 3.1 b, but with derivative w.r.t. quasi-Fermi level difference.

While deriving the diode equations, we assume that within the depletion region, the difference between the electron and hole quasi-Fermi levels, $E_{Fn} - E_{Fp} = qV_A$. In figure 3.2.a, $E_{Fn} - E_{Fp}$ at $x=0$, is plotted vs V_A . The quasi-Fermi levels are obtained in the program from n and p using

equations: $E_{F,n} = E_i + k_B T \ln(n / n_i)$ and $E_{F,p} = E_i - k_B T \ln(p / n_i)$. One can see the saturation at high voltages in figure 3.2a. Fig 3.2b plots the derivative of log of current vs quasi-Fermi level difference. We can see a plateau corresponding to slope of about 0.6 which may be assigned to high level injection.

For the above simulations maximum voltage applied is 1.2 V which is much higher than the built-in voltage of 0.84 V. The voltage variation across the pn junction with applied voltage of 1.2 V is plotted in the figure 3.3. We can see an upward voltage step across the depletion region.

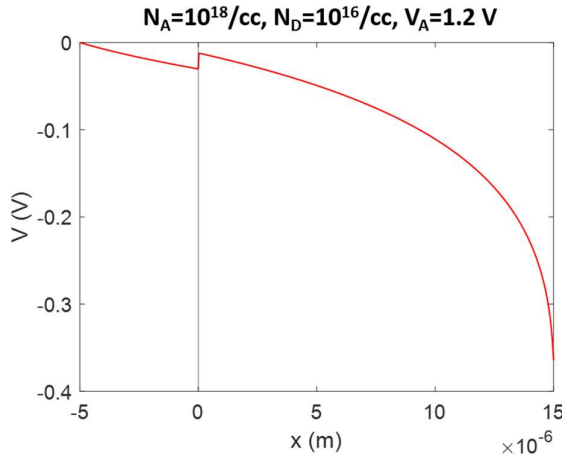


Figure 3.3 Voltage profile inside pn junction

The figure 3.4a shows the electron, hole and total current densities as a function of position for applied voltage of 0.65 V. The numerical results are compared with ideal diode theory. The inset shows the total current density (multiplied by 10^{-4}) as a function of position. We can see that the current density is almost constant as expected. Figure 3.4b shows hole current density decomposed into drift and diffusion components. On the n-side, outside the depletion region, we can see that J_p is almost equal to diffusion current density, as expected from ideal diode theory. The inset shows the current densities inside the depletion region. Both the drift and diffusion components have huge magnitudes (the y axis scale of inset is 10^9 !). The drift component is negative as the electric field inside depletion region of pn junction is negative. The diffusion component positive: The electron concentration would increase when going from p to n side giving rise to positive diffusion current. (Exercise: Obtain the drift and diffusion components of current density in a pn junction in equilibrium. You can also estimate these components from the potential profile obtained from the depletion approximation.)

Exercise on Solar cell IV: Consider a pn junction with light incident on the depletion region. The effect of light can be taken into account by introducing a generation term in the continuity equations. Simulate I-V under these conditions.

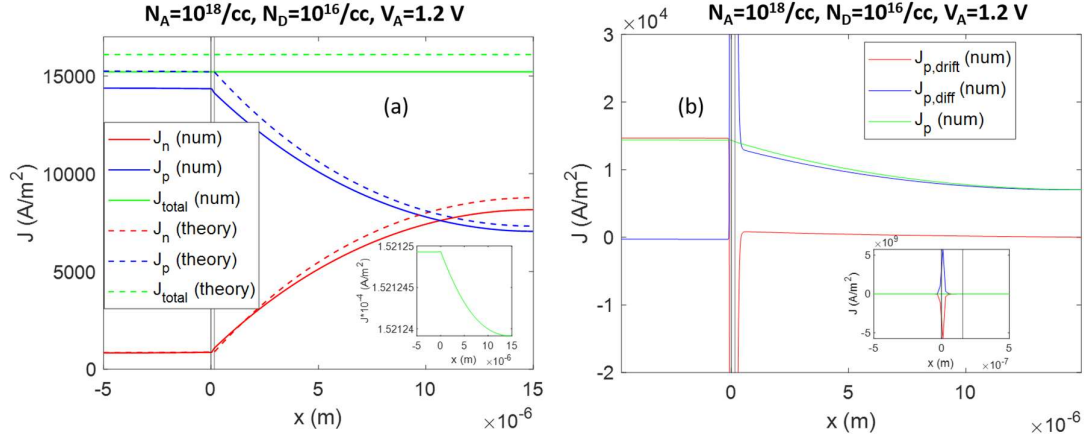


Figure 3.4(a) Current density vs in pn junction (b) decomposition of hole current density into drift and diffusion components.

4. Calculation of small signal admittance of pn junction:

The admittance of pn junction is calculated as follows. First get the n, p and ϕ for the applied dc bias. When a small ac voltage ($\delta V_A \exp(-i\omega t)$) is superimposed on the dc bias, we assume a solution of the following form satisfies equation 3.14.

$$\left. \begin{aligned} \tilde{n}_i &= \tilde{n}_{i,dc} + \delta \tilde{n}_i \exp(-i\omega t) \\ \tilde{p}_i &= \tilde{p}_{i,dc} + \delta \tilde{p}_i \exp(-i\omega t) \\ \tilde{\phi}_i &= \tilde{\phi}_{i,dc} + \delta \tilde{\phi}_i \exp(-i\omega t) \end{aligned} \right\} \text{--- (4.1)}$$

We then separate dc and ac parts of the equations. We also need boundary conditions to solve the equations. They are given by:

$$\left. \begin{aligned} \tilde{n}_L &= n_{0L} / N_1, \tilde{p}_L = p_{0L} / N_1 \\ \tilde{n}_R &= n_{0R} / N_2, \tilde{p}_R = p_{0R} / N_2 \\ \tilde{\phi}_L &= 0, \tilde{\phi}_R = [V_{bi} - (V_A + \delta V_a)] / \phi_N \end{aligned} \right\} \text{--- (4.2)}$$

The boundary conditions can also be separated out for the dc and ac part. The boundary condition for dc part is same as equation 3.15, and for ac part the b.c. are:

$$\left. \begin{aligned} \delta \tilde{n}_L &= 0, \delta \tilde{p}_L = 0 \\ \delta \tilde{n}_R &= 0, \delta \tilde{p}_R = 0 \\ \delta \tilde{\phi}_L &= 0, \delta \tilde{\phi}_R = -\delta V_A / \phi_N \end{aligned} \right\} \text{--- (4.3)}$$

The solution for the dc part with boundary conditions is already obtained. The equations for the ac part are given by:

$$\left[\frac{\partial f_i}{\partial \tilde{n}} \delta \tilde{n} + \frac{\partial f_i}{\partial \tilde{p}} \delta \tilde{p} + \frac{\partial f_i}{\partial \tilde{\phi}} \delta \tilde{\phi} \right] + \frac{\partial f_i}{\partial \tilde{\phi}_R} \delta \tilde{\phi}_R = \tau_1 (-i\omega \tilde{n}_i), \text{ (electron block)} \quad (4.4a)$$

$$\left[\frac{\partial f_i}{\partial \tilde{n}} \delta \tilde{n} + \frac{\partial f_i}{\partial \tilde{p}} \delta \tilde{p} + \frac{\partial f_i}{\partial \tilde{\phi}} \delta \tilde{\phi} \right] + \frac{\partial f_i}{\partial \tilde{\phi}_R} \delta \tilde{\phi}_R = \tau_2 (-i\omega \tilde{p}_i), \text{ (hole block)} \quad (4.4b)$$

$$\left[\frac{\partial f_i}{\partial \tilde{n}} \delta \tilde{n} + \frac{\partial f_i}{\partial \tilde{p}} \delta \tilde{p} + \frac{\partial f_i}{\partial \tilde{\phi}} \delta \tilde{\phi} \right] + \frac{\partial f_i}{\partial \tilde{\phi}_R} \delta \tilde{\phi}_R = 0, \text{ (Poisson block)} \quad (4.4c)$$

Note that the above equations have derivative w.r.t. $\tilde{\phi}_R$ as $\delta \tilde{\phi}_R$ is the only non-zero boundary condition. This term is present only in the last equations of each block.

The above equations can be compactly written as:

$$\underbrace{\left\{ \begin{pmatrix} \frac{\partial f_e}{\partial \tilde{n}} & \frac{\partial f_e}{\partial \tilde{p}} & \frac{\partial f_e}{\partial \tilde{\phi}} \\ \frac{\partial f_p}{\partial \tilde{n}} & \frac{\partial f_p}{\partial \tilde{p}} & \frac{\partial f_p}{\partial \tilde{\phi}} \\ \frac{\partial f_{poi}}{\partial \tilde{n}} & \frac{\partial f_{poi}}{\partial \tilde{p}} & \frac{\partial f_{poi}}{\partial \tilde{\phi}} \end{pmatrix} \right\}}_{\mathcal{J}} + i\omega \begin{pmatrix} \tau_1 I & 0 & 0 \\ 0 & \tau_2 I & 0 \\ 0 & 0 & 0I \end{pmatrix} \begin{pmatrix} \delta \tilde{n} \\ \delta \tilde{p} \\ \delta \tilde{\phi} \end{pmatrix} = - \begin{pmatrix} \frac{\partial f_e}{\partial \tilde{\phi}_R} \\ \frac{\partial f_p}{\partial \tilde{\phi}_R} \\ \frac{\partial f_{poi}}{\partial \tilde{\phi}_R} \end{pmatrix} \delta \tilde{\phi}_R \quad (4.5)$$

Thus, the solution for $(\delta \tilde{n}, \delta \tilde{p}, \delta \tilde{\phi})$ can be obtained by inverting the matrix on LHS or better by using back slash operator of MATLAB. This is a single shot procedure without any iterations. The Jacobian in the above equation is the same as in equation 3.16.

Once $\delta \tilde{n}, \delta \tilde{p}$ and $\delta \tilde{\phi}$ are obtained from equation 4.5, the electron and hole ac current densities are calculated as:

$$\delta J_{n,i+1/2} = \frac{\partial J_{n,i+1/2}}{\partial \tilde{n}} \delta \tilde{n} + \frac{\partial J_{n,i+1/2}}{\partial \tilde{\phi}} \delta \tilde{\phi} + \frac{\partial J_{n,i+1/2}}{\partial \tilde{\phi}_R} \delta \tilde{\phi}_R \quad (4.6a)$$

$$\delta J_{p,i+1/2} = \frac{\partial J_{p,i+1/2}}{\partial \tilde{p}} \delta \tilde{p} + \frac{\partial J_{p,i+1/2}}{\partial \tilde{\phi}} \delta \tilde{\phi} + \frac{\partial J_{p,i+1/2}}{\partial \tilde{\phi}_R} \delta \tilde{\phi}_R \quad (4.6b)$$

The derivatives of J_n and J_p can be obtained from equations 3.23 a and b. The total current density is given by adding displacement current density (J_D) to electron and hole current densities.

$$\left. \begin{aligned} \delta J &= \delta J_n + \delta J_p + J_D \\ \text{where } J_D &= \frac{\partial D}{\partial t} = \varepsilon(-i\omega \delta E) \end{aligned} \right\} \quad (4.7)$$

The complex admittance is then obtained from $Y = \delta J / \delta V_A$. Some of the simulation results of capacitance of p⁺n junction as a function of bias voltage (both FB and RB) for a fixed frequency are shown below. The simulation results are also compared with theory.

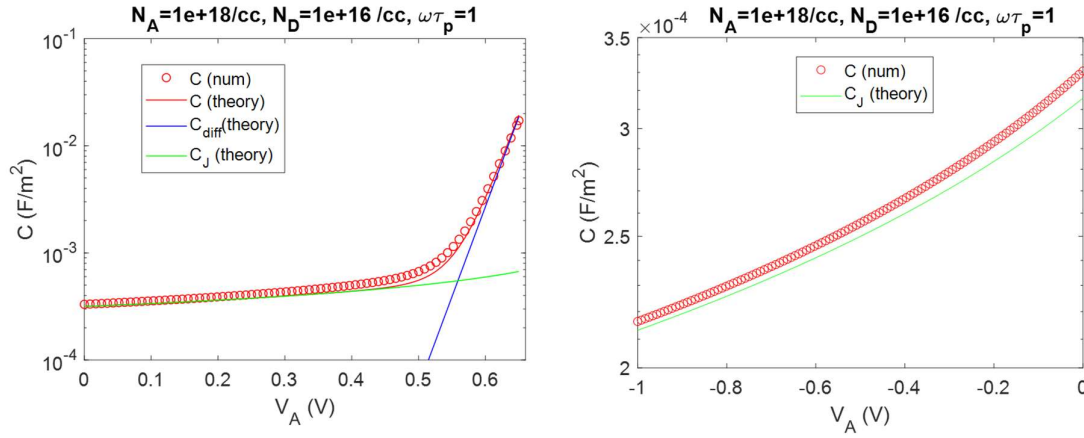


Figure 4.1

Capacitance and conductance as a function of frequency for a fixed applied voltage of 0.65 V are shown below:

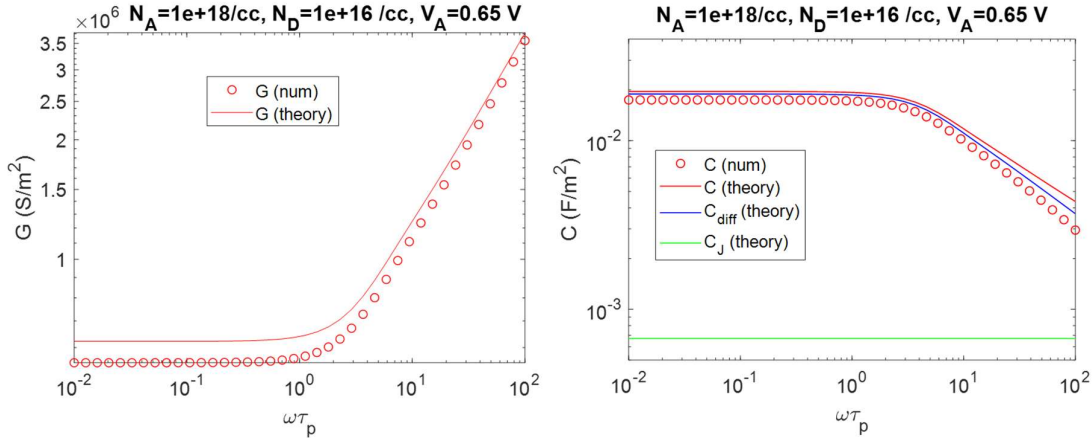


Figure 4.2

The ac current flow through the pn junction as a function of position obtained while calculating admittance are plotted in the figure below. The real part corresponds to the current in phase with applied voltage and imaginary part is 90° out of phase with voltage.

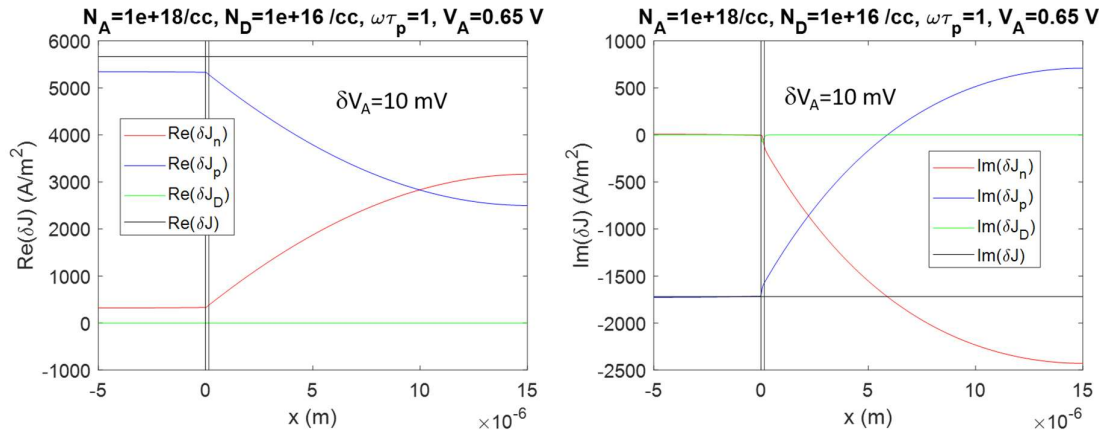


Figure 4.3

We can see that both the real and imaginary parts of the current density are constant as a function of position, as should be. The imaginary part of displacement current density shows a dip in the depletion region. This is due to the oscillation of electric field in the depletion region in-phase with the applied voltage.

5. Calculation of capacitance of MOS-capacitor

We are going to consider ideal MOS capacitor here. The procedure to find out the small signal impedance is almost same as what was done for pn junction. First get the n, p and ϕ for the applied dc gate voltage bias and then a small ac voltage is superimposed on the dc bias. In the case of MOS, when dc gate voltage is applied, semiconductor is in equilibrium (Fermi level is constant) and we need to solve only the Poisson equation to get the potential as a function of position. The electron and hole density are then calculated from potential using equation 1.3.

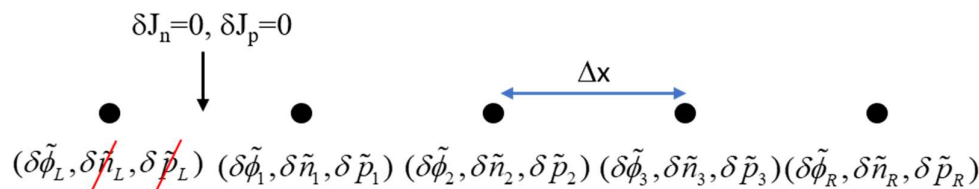
If we know the surface potential, we can solve differential equation 1.15 (degenerate case) or 1.8 (non-degenerate case) numerically to get potential as a function of x. This procedure has been followed to calculate the potential inside pn junction in equilibrium. In the case of MOS capacitor, the surface potential can be obtained from the gate voltage using the relation:

$$V_G = \phi_s + \frac{K_s}{K_{ox}} t_{ox} E_s \quad \text{--- (5.1)}$$

The surface electric field (E_s) is related to surface potential (ϕ_s) through equation 1.8 in non-degenerate case or equation 1.15 in degenerate case. ϕ_s can be found out numerically from above equation using fzero function of MATLAB.

Here we consider non-degenerate approximation to find out capacitance. Degenerate case is treated in the Appendix. After we have found out the potential, density of electrons and holes for an applied gate voltage (equilibrium solution), we can find out capacitance by superposing a small ac gate voltage, $\delta V_G \exp(-i\omega t)$ and finding out the ac current flow through MOS capacitor. As before, we need to solve current density continuity equations for electrons and holes, and Poisson equation to get current density.

In the MOS case, the RHS of semiconductor is taken as connected to a grounded contact, which implies $\delta\phi_R=0$, $\delta n_R=0$ and $\delta p_R=0$. The surface potential ($\phi_L = \phi_s$) oscillates when the gate voltage oscillates. Note that the LHS of semiconductor is not connected to a contact and $\delta n_L =$ and δp_L are not 0. We do not know δn_L and δp_L directly, but rather we have a different boundary condition here. The discretization and boundary conditions are shown schematically below:



We need to put the boundary condition that electron and hole current densities are zero at the left side, as electrons and holes cannot flow into the insulator. (Note that there is however a nonzero displacement current density at the interface.) The terms involving δn_L and δp_L appear in the first equation of electron and hole block respectively. These terms can be eliminated by using boundary condition that J_n and J_p are 0 at the left boundary as follows (see equation 3.5a):

$$\begin{aligned}
 f_1 &\equiv \frac{1}{q} \frac{\tau_1}{N_1} \frac{J_{n,1.5} - J_{n,0.5}}{\Delta x} - R_n(\tilde{n}_1, \tilde{p}_1) \\
 - > \delta f_1 &\equiv \frac{1}{q} \frac{\tau_1}{N_1} \frac{\delta J_{n,1.5} - \delta J_{n,0.5}}{\Delta x} - \delta R_n(\tilde{n}_1, \tilde{p}_1). \text{ put } \delta J_{n,0.5} = 0 \\
 - > 1st \text{ eqn is :} \\
 \frac{\tau_1}{q N_1 \Delta x} & \left[\frac{\partial J_{n,1.5}}{\partial \tilde{n}_1} \delta \tilde{n}_1 + \frac{\partial J_{n,1.5}}{\partial \tilde{n}_2} \delta \tilde{n}_2 + \frac{\partial J_{n,1.5}}{\partial \tilde{\phi}_1} \delta \tilde{\phi}_1 + \frac{\partial J_{n,1.5}}{\partial \tilde{\phi}_2} \delta \tilde{\phi}_2 \right] - \frac{\partial R_n(1)}{\partial \tilde{n}_1} \delta \tilde{n}_1 - \frac{\partial R_n(1)}{\partial \tilde{p}_1} \delta \tilde{p}_1 = -i\omega \tau_1 \delta \tilde{n}_1 - (5.2)
 \end{aligned}$$

$\delta J_{1.5}$ depends only on n and ϕ at points 1 and 2. Thus δn_L is not required to be known. Same applies to the first equation of the hole block. From equation 3.5 b, we get:

$$\begin{aligned}
 f_{N+1} &\equiv -\frac{1}{q} \frac{\tau_2}{N_2} \frac{J_{p,1.5} - J_{p,0.5}}{\Delta x} - R_p(\tilde{n}_1, \tilde{p}_1) \\
 - > \delta f_{N+1} &\equiv -\frac{1}{q} \frac{\tau_2}{N_2} \frac{\delta J_{p,1.5} - \delta J_{p,0.5}}{\Delta x} - \delta R_p(\tilde{n}_1, \tilde{p}_1). \text{ put } \delta J_{p,0.5} = 0 \\
 - > N+1st \text{ eqn is :} \\
 -\frac{\tau_2}{q N_2 \Delta x} & \left[\frac{\partial J_{p,1.5}}{\partial \tilde{p}_1} \delta \tilde{p}_1 + \frac{\partial J_{p,1.5}}{\partial \tilde{p}_2} \delta \tilde{p}_2 + \frac{\partial J_{p,1.5}}{\partial \tilde{\phi}_1} \delta \tilde{\phi}_1 + \frac{\partial J_{p,1.5}}{\partial \tilde{\phi}_2} \delta \tilde{\phi}_2 \right] - \frac{\partial R_p(1)}{\partial \tilde{n}_1} \delta \tilde{n}_1 - \frac{\partial R_p(1)}{\partial \tilde{p}_1} \delta \tilde{p}_1 = -i\omega \tau_2 \delta \tilde{p}_1 - (5.3)
 \end{aligned}$$

We can write the equations to be solved as :

$$\underbrace{\left\{ \begin{pmatrix} \frac{\partial f_e}{\partial \tilde{n}} & \frac{\partial f_e}{\partial \tilde{p}} & \frac{\partial f_e}{\partial \tilde{\phi}} \\ \frac{\partial f_p}{\partial \tilde{n}} & \frac{\partial f_p}{\partial \tilde{p}} & \frac{\partial f_p}{\partial \tilde{\phi}} \\ \frac{\partial f_{poi}}{\partial \tilde{n}} & \frac{\partial f_{poi}}{\partial \tilde{p}} & \frac{\partial f_{poi}}{\partial \tilde{\phi}} \end{pmatrix} \right\}}_{\mathcal{J}'} + i\omega \begin{pmatrix} \tau_1 I & 0 & 0 \\ 0 & \tau_2 I & 0 \\ 0 & 0 & 0I \end{pmatrix} \begin{pmatrix} \delta \tilde{n} \\ \delta \tilde{p} \\ \delta \tilde{\phi} \end{pmatrix} = - \begin{pmatrix} \frac{\partial f_e}{\partial \tilde{\phi}_L} \\ \frac{\partial f_p}{\partial \tilde{\phi}_L} \\ \frac{\partial f_{poi}}{\partial \tilde{\phi}_L} \end{pmatrix} \delta \tilde{\phi}_L \dots (5.4)$$

These equations are of the same form as equation 4.5. The Jacobian now is a bit different as the first equations of electron and hole blocks are different. Once $\delta \tilde{n}, \delta \tilde{p}$ and $\delta \tilde{\phi}$ are obtained from equation 5.4, the ac current densities can be calculated from equations 4.6 and 4.7. The oscillating gate voltage is related to oscillating gate voltage as:

$$\begin{aligned}
 \delta V_G &= \delta \phi_L + \frac{K_S}{K_{ox}} t_{ox} \delta E_S = \delta \phi_L + \frac{K_S}{K_{ox}} t_{ox} \left(\frac{i \delta J_{D,S}}{K_S \epsilon_0 \omega} \right) \\
 &= \delta \phi_L + \left(\frac{i \delta J}{C_{ox} \omega} \right) - (5.5)
 \end{aligned}$$

In the last step we have replaced δJ_D at the surface by δJ , as the current is conserved and at the surface electron and hole currents are zero. This is important because for simulations at low frequencies and for gate voltage above threshold voltage, MATALB gives a warning that the solution may not be correct. In such cases, if you plot current density as a function of position, you can see that near the interface it is not really constant. We therefore take average current density away from the surface while using equation 5.5. The complex admittance is obtained from $Y = \delta J / \delta V_G$.

Some of the simulation results as a function of gate voltage for different frequencies are shown in figure 5.1. The capacitance is compared with low frequency capacitance expected from theory. ($E(\phi)$ function is defined in the code, and the low frequency capacitance can be easily calculated from it. This is left as an exercise.)

We can also see a frequency dependent conductance for gate voltages more than the threshold voltage. The non-zero conductance would imply that there is dissipation. Figuring out the source of dissipation is left as an exercise.

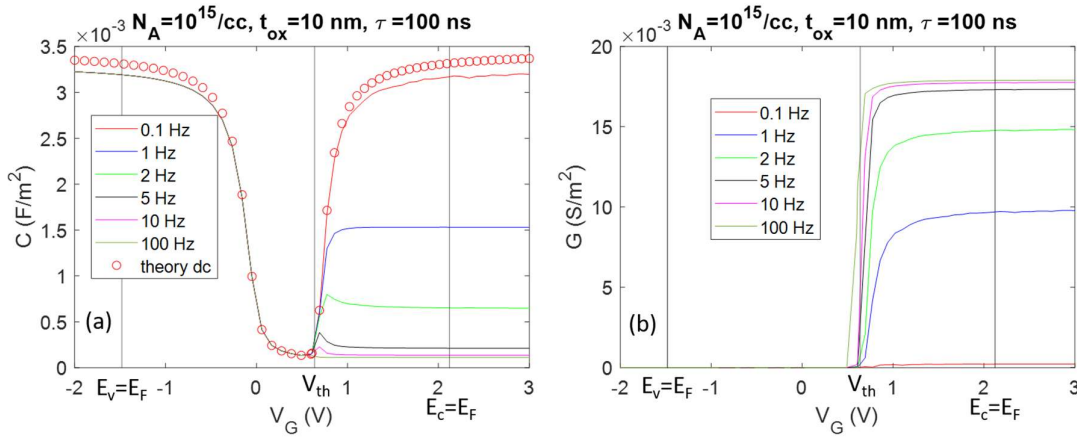


Figure 5.1(a) Capacitance (b) conductance as a function of gate voltage. The threshold voltage and gate voltages at which the conduction band and valence band touch the Fermi level are also shown.

The current density flowing the semiconductor on application of ac gate voltage is shown in figure 5.2. We can see that the total current density near the surface is a bit noisy, but constant away from the surface. MATALB gives a warning that the solution may not be correct. The numerical inaccuracies may result in the non-constant total current density near the interface.

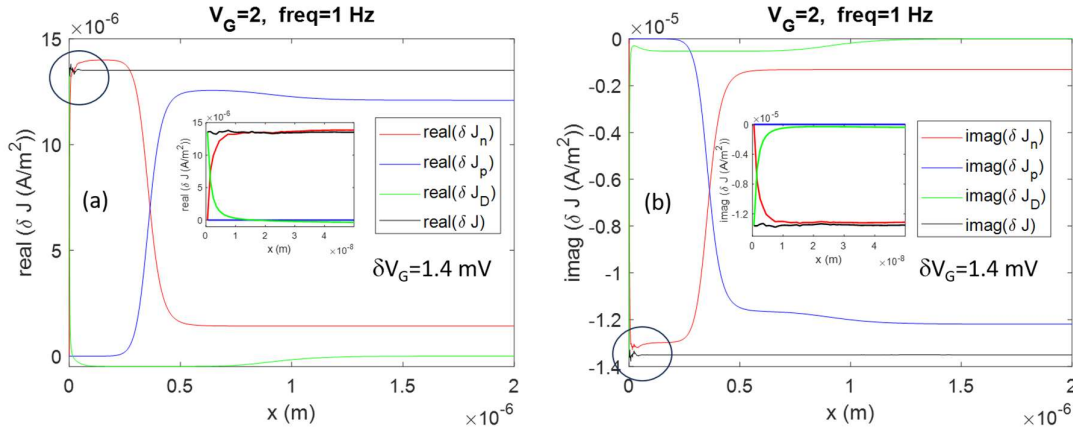


Figure 5.2 The (a) real and (b) imaginary current densities as a function of position in Si. The circle shows the part where current density is not flat. The inset shows zoomed figure near the surface.

6. Time dependent simulations

To understand the method that we will use for solving the time dependent continuity equations of semiconductor, let us first consider a simpler time dependent diffusion equation given below.

$$\frac{\partial n}{\partial t} = D \frac{\partial^2 n}{\partial x^2} \quad \text{--- (6.1)}$$

We want to find out $n(x, t > 0)$ if we are given $n(x, t = 0)$, by solving above equation numerically. We will now discretize the above equation in space and time. The RHS of this equation can be discretized as:

$$\frac{\partial^2 n}{\partial x^2} = D \frac{n_{i+1} - 2n_i + n_{i-1}}{(\Delta x)^2} \equiv f_n \quad \text{--- (6.2)}$$

We can write the time derivative of n as:

$$\frac{\partial n_i}{\partial t} = \frac{n_i(t + \Delta t) - n_i(t)}{\Delta t} \quad \text{--- (6.3)}$$

Thus, the equation to be solved can be written as:

$$\frac{\partial n}{\partial t} = f_n(t) \quad \text{--- (6.4a)}$$

$$\Rightarrow \frac{n_i(t + \Delta t) - n_i(t)}{\Delta t} = D \frac{n_{i+1}(t) - 2n_i(t) + n_{i-1}(t)}{(\Delta x)^2}$$

$$\Rightarrow n_i(t + \Delta t) = n_i(t) + D \frac{\Delta t}{(\Delta x)^2} [n_{i+1}(t) - 2n_i(t) + n_{i-1}(t)] \quad \text{--- (6.4b)}$$

This if we know n_i at time t , we can get n_i at time $t + \Delta t$ using equation 6.4b. This method is called as forward Euler method. However, this method can show numerical instability, and we will instead use Crank-Nicolson method, where we take average of $f_n(t)$ and $f_n(t + \Delta t)$ in equation 6.4a. We write:

$$\frac{\partial n}{\partial t} = \frac{1}{2} [f_n(t) + f_n(t + \Delta t)] \text{--- (6.5a)}$$

$$\Rightarrow \frac{n_i(t + \Delta t) - n_i(t)}{\Delta t} = \frac{D}{2} \left[\frac{n_{i+1}(t) - 2n_i(t) + n_{i-1}(t)}{(\Delta x)^2} + \frac{n_{i+1}(t + \Delta t) - 2n_i(t + \Delta t) + n_{i-1}(t + \Delta t)}{(\Delta x)^2} \right] \text{--- (6.5b)}$$

If we know n_i at time t , we can get n_i at time $t + \Delta t$ using equation 6.5b.

Let us now consider a non-degenerate semiconductor. The electron and hole current density continuity equations and Poisson equation that we need to solve are given by 3.1b and 3.1d. The discretized equations for the scaled variables are given by equations 3.10 and 3.11.

These equations are of the form:

$$\tau_1 \frac{\partial \tilde{n}}{\partial t} = f_n(\tilde{n}, \tilde{p}, \tilde{\phi}) \text{--- (6.6a)}$$

$$\tau_2 \frac{\partial \tilde{p}}{\partial t} = f_p(\tilde{n}, \tilde{p}, \tilde{\phi}) \text{--- (6.6b)}$$

$$0 = f_{poi}(\tilde{n}, \tilde{p}, \tilde{\phi}) \text{--- (6.6c)}$$

Now our problem is given $(\tilde{n}, \tilde{p}, \tilde{\phi})$ at time t , how do we get their values at time $t + \Delta t$. By using Crank-Nicolson method (equation 6.5 a), we can write:

$$\left. \begin{aligned} \tau_1 \frac{\tilde{n}_i(t + \Delta t) - \tilde{n}_i(t)}{\Delta t} &= \frac{1}{2} [f_n(\tilde{n}(t), \tilde{p}(t), \tilde{\phi}(t)) + f_n(\tilde{n}(t + \Delta t), \tilde{p}(t + \Delta t), \tilde{\phi}(t + \Delta t))] \\ \tau_2 \frac{\tilde{p}_i(t + \Delta t) - \tilde{p}_i(t)}{\Delta t} &= \frac{1}{2} [f_p(\tilde{n}(t), \tilde{p}(t), \tilde{\phi}(t)) + f_p(\tilde{n}(t + \Delta t), \tilde{p}(t + \Delta t), \tilde{\phi}(t + \Delta t))] \\ 0 &= \frac{1}{2} [f_{poi}(\tilde{n}(t), \tilde{p}(t), \tilde{\phi}(t)) + f_{poi}(\tilde{n}(t + \Delta t), \tilde{p}(t + \Delta t), \tilde{\phi}(t + \Delta t))] \end{aligned} \right\} \text{--- (6.7)}$$

These equations can be written in the form:

$$\begin{aligned} g_n(\tilde{n}(t + \Delta t), \tilde{p}(t + \Delta t), \tilde{\phi}(t + \Delta t)) &\equiv [f_n(\tilde{n}(t + \Delta t), \tilde{p}(t + \Delta t), \tilde{\phi}(t + \Delta t)) - 2\tau_1 \frac{\tilde{n}_i(t + \Delta t)}{\Delta t}] + [2\tau_1 \frac{\tilde{n}_i(t)}{\Delta t} + f_n(\tilde{n}(t), \tilde{p}(t), \tilde{\phi}(t))] = 0 \\ g_p(\tilde{n}(t + \Delta t), \tilde{p}(t + \Delta t), \tilde{\phi}(t + \Delta t)) &\equiv [f_p(\tilde{n}(t + \Delta t), \tilde{p}(t + \Delta t), \tilde{\phi}(t + \Delta t)) - 2\tau_2 \frac{\tilde{p}_i(t + \Delta t)}{\Delta t}] + [2\tau_2 \frac{\tilde{p}_i(t)}{\Delta t} + f_p(\tilde{n}(t), \tilde{p}(t), \tilde{\phi}(t))] = 0 \\ g_{poi}(\tilde{n}(t + \Delta t), \tilde{p}(t + \Delta t), \tilde{\phi}(t + \Delta t)) &\equiv [f_{poi}(\tilde{n}(t + \Delta t), \tilde{p}(t + \Delta t), \tilde{\phi}(t + \Delta t))] + [f_{poi}(\tilde{n}(t), \tilde{p}(t), \tilde{\phi}(t))] = 0 \\ &\text{--- (6.8a, b, c)} \end{aligned}$$

Note that in above equations, $(\tilde{n}(t + \Delta t), \tilde{p}(t + \Delta t), \tilde{\phi}(t + \Delta t))$ are three unknowns at each node. The solution at time t , $(\tilde{n}(t), \tilde{p}(t), \tilde{\phi}(t))$ is known. The functions f_n , f_p and f_{poi} are defined before. $f_{poi}(\tilde{n}(t), \tilde{p}(t), \tilde{\phi}(t))$ in above equation can be put as 0 (equation 6.8c).

The equations 6.8 can be solved using Newton-Raphson method as done before: we assume solution at time t as the initial guess and then obtain solution of the above equations iteratively. Thus, at each time step, we need to solve the above equations iteratively. We can use either Newton or Gummel method.

For spatial discretization, Δx should be chosen less than the Debye length. For choosing time step (Δt), there are two time scales that we should consider. The dielectric relaxation time is given by (Taur and Ning, equation 2.70):

$$t_{dr} = \frac{\epsilon}{qN\mu_{majority}} \text{ --- (6.9)}$$

Where N is the doping concentration and $\mu_{majority}$ is the mobility of majority carriers. For diffusion equation of minority carriers, the time scale is given by: $\tau = (\Delta x)^2 / 2D$. Putting $\Delta x = \text{Debye length}$ (exercise below equation 1.8), and $D = \mu V_{therm}$, we get

$$\tau = \frac{L_D^2}{2D_{minority}} = \frac{\epsilon}{2qN\mu_{minority}} \text{ --- (6.10)}$$

The time step should be chosen less than t_{dr} and τ .

A: Simulation of Haynes-Shockley experiment

Haynes-Shockley experiment is a classic experiment which demonstrates the transport of carriers in a semiconductor by drift and diffusion. The experiment also demonstrates effect of recombination.

Consider a n-type semiconductor with dc voltage applied across it as shown in figure below. This creates a uniform electric field along x direction inside the semiconductor. If electron-hole pairs are created in a small region of the semiconductor, the holes (minority carriers) drift to right side due to the external electric field and also undergo diffusion. The extra electrons (majority carriers) follow the holes due to the internal electric field. The carriers can be injected into the semiconductor electrically (as done in the original Haynes-Shockley experiment) or optically. A pulse of light can be used to create electron-hole pairs. The transport of injected carriers can be detected by using a rectifying contact.

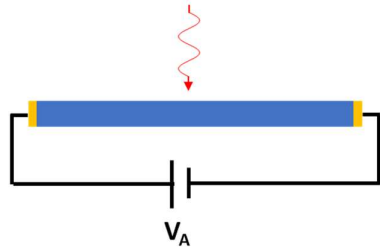


Figure 6.1 Schematic of the experiment. Semiconductor is biased with a dc voltage and a light pulse is incident to create electron-hole pairs locally. The minority carriers undergo drift and diffusion.

Typical experiments are done on large samples, few mm in length. However, we here simulate 1d semiconductors with length of 5-10 μm . The Debye length scale in a semiconductor is given

by: $\sqrt{\epsilon_0 K_S k_B T / q^2 (p_0 + n_0)}$ (see exercise problem below equation 1.8). For a doping of $10^{16}/\text{cc}$, this is about 40 nm for Si. That is why we cannot simulate Si with long length.

The simulation results are shown below. N-type Si ($N_D=10^{16}/\text{cc}$) with 10 μm length and applied voltage of 0.3 V is considered. A light pulse with Gaussian intensity is first incident on the semiconductor for 1 ps. This creates electron-hole pairs. This is simulated by adding generation term to the continuity equations 3.1b. The resultant excess carriers are shown by the red curves in figure 6.2 a and b. The electron and hole concentrations, 1ns after the pulse is switched off are shown in the figure 6.2 by blue curve. The equations 6.8 which need to be solved iteratively are solved by Newton's method. (We find that Gummel method is quite efficient for solving pn junction current-voltage characteristics, however for this simulation Newton's method is more efficient.)

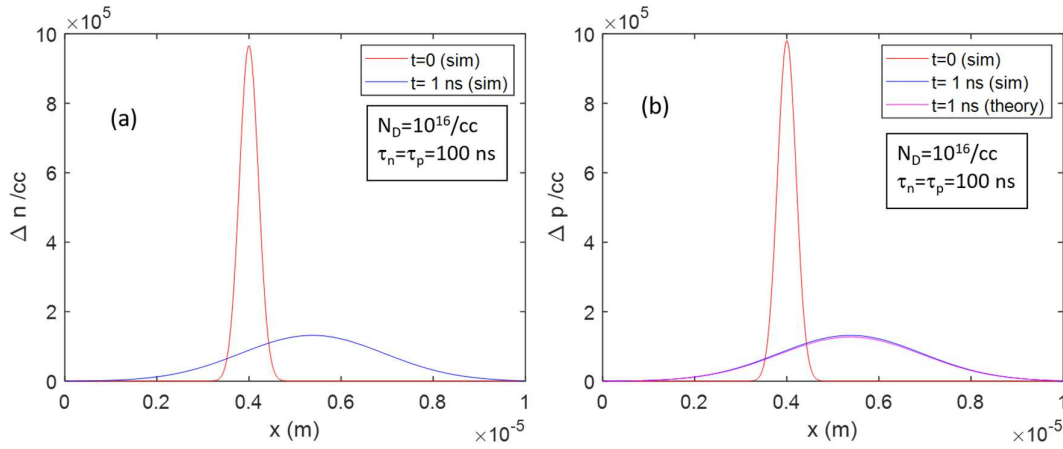


Figure 6.2

Assuming the initial hole (minority carrier) concentration is Gaussian in space, the expected hole concentration from the solution of minority carrier diffusion equation is given by:

$$\Delta p(x,t) = A \exp(-t / \tau_p) \frac{1}{\sigma \sqrt{2\pi}} \exp\left[-\frac{(x-x_{avg})^2}{2\sigma^2}\right] \quad \left. \vphantom{\Delta p(x,t)} \right\} \text{--- (6.11)}$$

where $x_{avg} = x_0 + \mu_p E t$, $\sigma = \sqrt{\sigma_0^2 + 2D_p t}$

The magenta curve in figure 6.2b is obtained using the above equation. The drift and diffusion hole current density at 1ns is shown in figure 6.3.

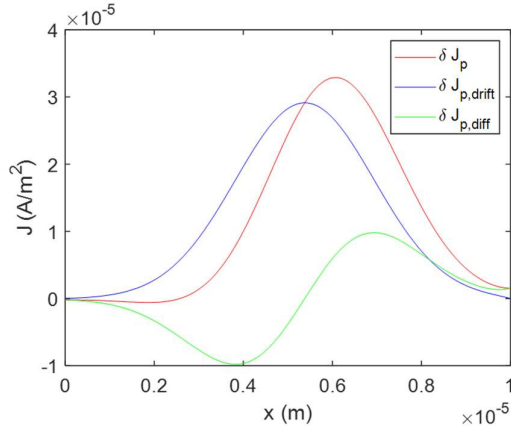


Figure 6.3 The drift and diffusion components of hole current density at $t=1\text{ ns}$.

In this example, the simulation time is much less than τ_p and it is difficult to see the effect of recombination (i.e. $\exp(-t/\tau_p) \sim 1$). To see this effect, you can take τ_p to be small, say 10 ns, and decrease the mobilities so that the value of Δt is more. You can also decrease the semiconductor length to make the calculation faster. This simulation is left as an exercise.

Appendix A: Derivation of Scharfetter-Gummel discretization Scheme:

If we know the electron and hole densities and voltage at node i and $i+1$ i.e. if know (n_i, p_i, ϕ_i) and $(n_{i+1}, p_{i+1}, \phi_{i+1})$, how do we find out their values in between? We can assume that voltage varies linearly between points x_i and x_{i+1} . n and p depend exponentially on voltage in equilibrium. They are expected to vary quite rapidly between x_i to x_{i+1} even in non-equilibrium case i.e. we should not take linear variation of n and p between x_i to x_{i+1} . To find out how n varies, we assume J_n is constant between x_i to x_{i+1} . We now need to find out $n(x)$ between x_i to x_{i+1} , such that J_n is constant. Since potential varies linearly, E and hence μ are constant between x_i to x_{i+1} . The steps to get $n(x)$ and J_n , from (n_i, p_i, ϕ_i) and $(n_{i+1}, p_{i+1}, \phi_{i+1})$ are given below:

we have $q(\mu_n n E + D_n \frac{\partial n}{\partial x}) = J_n$: eqn 21a

take J_n , μ_n , D_n and E as constant over interval i to $i+1$

multiply both sides by integrating factor: $\exp[(x - x_i) \frac{\mu_n E}{D_n}]$

$$LHS = q D_n \frac{\partial}{\partial x} \{ n \exp[(x - x_i) \frac{\mu_n E}{D_n}] \}$$

$$\text{thus we get, } q D_n \frac{\partial}{\partial x} \{ n \exp[(x - x_i) \frac{\mu_n E}{D_n}] \} = J_{n,i+1/2} \exp[(x - x_i) \frac{\mu_n E}{D_n}]$$

integrate above eqn from x_i to x

$$- > q D_n \{ n \exp[(x - x_i) \frac{\mu_n E}{D_n}] \}_{x_i}^x = J_{n,i+1/2} \frac{D_n}{\mu_n E} \{ \exp[(x - x_i) \frac{\mu_n E}{D_n}] \}_{x_i}^x$$

$$- > q \mu_n E \{ n \exp[(x - x_i) \frac{\mu_n E}{D_n}] - n_i \} = J_{n,i+1/2} \{ \exp[(x - x_i) \frac{\mu_n E}{D_n}] - 1 \} \text{ --- (A1)}$$

put $x = x_{i+1}$ in above eqn

$$- > J_{n,i+1/2} = q \mu_n E \frac{\{ n(x_{i+1}) \exp[\Delta x \frac{\mu_n E}{D_n}] - n(x_i) \}}{\{ \exp[\Delta x \frac{\mu_n E}{D_n}] - 1 \}} \text{ --- (A2)}$$

We have thus obtained expression for J_n if we are given (n_i, p_i, ϕ_i) and $(n_{i+1}, p_{i+1}, \phi_{i+1})$. We can now insert equation A2 into A1 to eliminate $J_{n,i+1/2}$ and get $n(x)$.

$$\text{we get, } q \mu_n E \{ n \exp[(x - x_i) \frac{\mu_n E}{D_n}] - n_i \} = q \mu_n E \frac{\{ n(x_{i+1}) \exp[\Delta x \frac{\mu_n E}{D_n}] - n(x_i) \}}{\{ \exp[\Delta x \frac{\mu_n E}{D_n}] - 1 \}} \{ \exp[(x - x_i) \frac{\mu_n E}{D_n}] - 1 \}$$

after simplification,

$$n(x) = \frac{n_{i+1} \{ 1 - \exp[-(x - x_i) \frac{\mu_n E}{D_n}] \} + n_i \{ \exp[-(x - x_i) \frac{\mu_n E}{D_n}] - \exp(-\Delta x \frac{\mu_n E}{D_n}) \}}{[1 - \exp(-\Delta x \frac{\mu_n E}{D_n})]} \text{ --- (A3)}$$

You can check that if we put $x=x_i$ in above expression, we get $n=n_i$ and if we put $x=x_{i+1}$, we get $n=n_{i+1}$. We now write equations A2 and A3 in a simplified way:

use : $\frac{D_n}{\mu_n} = V_{therm}$ (Einstein relation)

define, $U = \frac{V}{V_{therm}} \rightarrow E = -V_{therm} \frac{U_{i+1} - U_i}{\Delta x} = -V_{ther} \frac{\Delta U}{\Delta x}$

$\rightarrow J_{n,i+1/2} = qD_n \frac{1}{\Delta x} \frac{\Delta U [n_{i+1} - n_i \exp(\Delta U)]}{[\exp(\Delta U) - 1]} \dots (A4)$

and $n(x) = \frac{n_{i+1} \{1 - \exp[(x - x_i) \frac{\Delta U}{\Delta x}]\} + n_i \{\exp[(x - x_i) \frac{\Delta U}{\Delta x}] - \exp(\Delta U)\}}{[1 - \exp(\Delta U)]} \dots (A5)$

define Bernoulli function, $B(z) = \frac{z}{e^z - 1}$

use : $\frac{ze^z}{e^z - 1} = \frac{z}{1 - e^{-z}} = \frac{-z}{e^{-z} - 1} = B(-z)$

$\Rightarrow J_{n,i+1/2} = q \frac{1}{\Delta x} D_{n,i+1/2} [n_{i+1} B(\Delta U) - n_i B(-\Delta U)] \dots (A6)$

In equation A6 we have indicated that value of D_n is at point $i+1/2$. Let us now evaluate drift and diffusion current densities at $x=(x_{i+1}-x_i)/2$ using above expression for n (equation A5).

$J_{n,drift,i+1/2} = q\mu_n E n_{i+1/2}$
 $= q\mu_n E \frac{n_{i+1} \{1 - \exp[\frac{\Delta U}{2}]\} + n_i \{\exp[\frac{\Delta U}{2}] - \exp(\Delta U)\}}{[1 - \exp(\Delta U)]}$

after simplification

$J_{n,drift,i+1/2} = -qD_n \frac{\Delta U}{\Delta x} \frac{n_{i+1} + n_i \exp(\frac{\Delta U}{2})}{[1 - \exp(\Delta U)]} \{1 - \exp[\frac{\Delta U}{2}]\}$
 $= qD_{n,i+1/2} \frac{B(\Delta U)}{\Delta x} \{n_{i+1} + n_i \exp(\frac{\Delta U}{2})\} \{1 - \exp[\frac{\Delta U}{2}]\} \dots (A7)$

Let us now evaluate diffusion current density:

$$\text{From eqn A4, } \frac{dn}{dx} = \frac{\Delta U}{\Delta x} \frac{n_{i+1} \{-\exp[(x-x_i) \frac{\Delta U}{\Delta x}]\} + n_i \{\exp[(x-x_i) \frac{\Delta U}{\Delta x}]\}}{[1 - \exp(\Delta U)]}$$

$$J_{n,diff,i+1/2} = q D_n \left. \frac{\partial n}{\partial x} \right|_{i+1/2}$$

$$= q D_n \frac{\Delta U}{\Delta x} \frac{n_{i+1} \{-\exp[\frac{\Delta U}{2}]\} + n_i \{\exp(\frac{\Delta U}{2})\}}{[1 - \exp(\Delta U)]}$$

after simplification,

$$J_{n,diff,i+1/2} = q D_n \frac{\Delta U}{\Delta x} \frac{-n_{i+1} + n_i}{[1 - \exp(\Delta U)]} \exp(\frac{\Delta U}{2})$$

$$= q D_{n,i+1/2} \frac{B(\Delta U)}{\Delta x} (n_{i+1} - n_i) \exp(\frac{\Delta U}{2}) \dots (A8)$$

You can verify from equations A6, A7 and A8 that $J_n = J_{n,drift} + J_{n,diff}$. Equations for hole current density can be obtained by noting that the sign of D_p is negative in expression for current density (equation 21). The final equations obtained are given below:

$$J_{p,i+1/2} = q \mu_p E \frac{\{p(x_{i+1}) \exp[\Delta x \frac{\mu_p E}{-D_p}] - p(x_i)\}}{\{\exp[\Delta x \frac{\mu_p E}{-D_p}] - 1\}}$$

$$J_{p,i+1/2} = -q \frac{1}{\Delta x} D_{p,i+1/2} [p_{i+1} B(-\Delta U) - p_i B(\Delta U)] \dots (A9)$$

$$p(x) = \frac{p_{i+1} \{1 - \exp[(x-x_i) \frac{-\Delta U}{\Delta x}]\} + p_i \{\exp[(x-x_i) \frac{-\Delta U}{\Delta x}] - \exp(-\Delta U)\}}{[1 - \exp(-\Delta U)]} \dots (A10)$$

$$J_{p,drift,i+1/2} = -q D_{p,i+1/2} \frac{B(-\Delta U)}{\Delta x} \{p_{i+1} + p_i \exp(-\frac{\Delta U}{2})\} \{1 - \exp(-\frac{\Delta U}{2})\} \dots (A11)$$

$$J_{p,diff,i+1/2} = q D_{p,i+1/2} \frac{B(-\Delta U)}{\Delta x} (p_i - p_{i+1}) \exp(-\frac{\Delta U}{2}) \dots (A12)$$

Appendix B: Equations for Degenerate case

In the non-degenerate case, we had 3N unknowns (n,p,φ) at each node and we had 3N equations: electron current density continuity, hole current density continuity and Poisson equations. In the degenerate case there are 5N unknowns including N_D^+ and N_A^- at each node. That means we need two more continuity equations for ionized impurities.

We are going to use the degenerate case solution to find out the MOS capacitance i.e. we are going to consider only small deviations from equilibrium. We write below a set of plausible equations.

$$\frac{\partial N_D^+}{\partial t} = \frac{N_D^+ - N_{D^+eq}(\phi)}{t_1} \quad \text{--- (B1)}, \text{ where } N_{D^+eq}(\phi) = \frac{N_D}{1 + g_D \exp\left[\left(\frac{E_{F,eq} - E_d + q\phi}{k_B T}\right)\right]}$$

$$\frac{\partial N_A^-}{\partial t} = \frac{N_A^- - N_{A^-eq}(\phi)}{t_2} \quad \text{--- (B2)}, \text{ where } N_{A^-eq}(\phi) = \frac{N_A}{1 + g_A \exp\left[\left(\frac{E_a - E_{F,eq} - q\phi}{k_B T}\right)\right]}$$

$$\frac{\partial n}{\partial t} = \frac{1}{q} \frac{dJ_n}{dx} - R + \frac{\partial N_D^+}{\partial t} \quad \text{--- (B3)}$$

$$\frac{\partial p}{\partial t} = -\frac{1}{q} \frac{dJ_p}{dx} - R + \frac{\partial N_A^-}{\partial t} \quad \text{--- (B4)}$$

$$\text{where } R = \frac{np - n_0 p_0}{\tau_p(n + n_i) + \tau_n(p + n_i)}$$

$$\frac{d^2 \phi}{dx^2} = -\frac{\rho}{\epsilon_0 K_s} \quad \text{--- (B5)}, \text{ where } \rho = q(p - n + N_D^+ - N_A^-)$$

We have essentially assumed the form of equations B1 and B2. These two equations essentially try to make concentrations of ionized impurities equal to what one would get if the system was in equilibrium with given value of potential.

New terms are added to the electron and hole current density continuity equations. You can verify that $\partial \rho / \partial t = -\partial J / \partial x$ is satisfied from equations B3 and B4. (However, this is not the only way to conserve current. e.g. we could have added $-\partial N_A^- / \partial t$ to equation B3 and $-\partial N_D^+ / \partial t$ to equation B4 and still satisfied current conservation.) Physically what equation B3 means is that the density of electrons in a volume element can change because of three factors viz. 1) electron current entering and exiting is different 2) generation and recombination of electrons with holes 3) electrons making transition to donor levels or electron from donor levels jumping to the conduction band. In principle, the electrons from the conduction band can also make transition to acceptor levels or electron from acceptor level jump to conduction band, however these processes are neglected as energy of conduction band edge is closer to donor levels and far away from acceptor levels. Same argument applies to equation B4.

We further assume that the time constants, t_1 and t_2 are very small. These conditions imply that:

$$N_D^+ \approx N_{D^+eq}(\phi) \quad \text{--- (B1')}$$

$$N_A^- \approx N_{A^-eq}(\phi) \quad \text{--- (B2')}$$

With these conditions we have only 3 unknowns (n, p, ϕ) and the three equations that we need to solve are given below:

$$\frac{\partial n}{\partial t} = \frac{1}{q} \frac{dJ_n}{dx} - R + \frac{dN_{D_{eq}}^+}{d\phi} \frac{\partial \phi}{\partial t} \quad \text{--- (B3')}$$

$$\frac{\partial p}{\partial t} = -\frac{1}{q} \frac{dJ_p}{dx} - R + \frac{dN_{A_{eq}}^-}{d\phi} \frac{\partial \phi}{\partial t} \quad \text{--- (B4')}$$

$$\frac{d^2 \phi}{dx^2} = -\frac{\rho}{\epsilon_0 K_s} \quad \text{--- (B5') , where } \rho = q [p - n + N_{D_{eq}}^+(\phi) - N_{A_{eq}}^-(\phi)]$$

The expression derivatives of N_D^+ and N_A^- are given by equation 2.9. Further note that in the non-degenerate region, these derivatives are approximately 0 and we recover the usual equations used before.

Appendix C: Discretization of electron and hole current densities in the degenerate case

In the degenerate case, the Einstein relations are modified as (See Pierret eq. 6.43):

$$\left. \begin{aligned} \frac{D_n}{\mu_n} &= \frac{K_B T}{q} \frac{\mathcal{F}_{1/2}(\eta_c)}{\mathcal{F}_{-1/2}(\eta_c)}, \text{ where } \eta_c = \frac{E_F - E_C + q\phi}{k_B T} \\ \frac{D_p}{\mu_p} &= \frac{K_B T}{q} \frac{\mathcal{F}_{1/2}(\eta_v)}{\mathcal{F}_{-1/2}(\eta_v)}, \text{ where } \eta_v = \frac{E_V - E_F - q\phi}{k_B T} \end{aligned} \right\} \quad \text{--- (C1)}$$

We can assume that μ is determined by the electric field and D is then can be obtained from above modified Einstein relations. Equation A2 was obtained assuming that D_n is constant over the range from x_i to x_{i+1} . However, in non-degenerate case, the ratio of D/μ depends on x and is determined by the equilibrium $\phi(x)$. Thus, D is a function of x and can vary over the range x_i to x_{i+1} . We here neglect this variation, and write equation A2 with the ratio D/μ taken as average of D/μ at points x_i and x_{i+1} . We can do a bit better by starting from equation A2 and taking the value of D/μ so that the current as given by equation A2 is zero in equilibrium. (Equation A2 is what we are going to use in the code for calculating current.)

$$\begin{aligned} \text{put } n_{i+1,eq} \exp\left[\Delta x \frac{\mu_n E}{D_n}\right] &= n_{i,eq} \\ \rightarrow \exp\left[\frac{\mu_n}{D_n}(\phi_{i,eq} - \phi_{i+1,eq})\right] &= \frac{\mathcal{F}_{1/2}\left(\frac{E_F - E_C + q\phi_{i,eq}}{k_B T}\right)}{\mathcal{F}_{1/2}\left(\frac{E_F - E_C + q\phi_{i+1,eq}}{k_B T}\right)} \\ \left(\frac{\mu_n}{D_n}\right)_{i+1/2} &= \frac{1}{(\phi_{i,eq} - \phi_{i+1,eq})} \ln\left[\frac{\mathcal{F}_{1/2}\left(\frac{E_F - E_C + q\phi_{i,eq}}{k_B T}\right)}{\mathcal{F}_{1/2}\left(\frac{E_F - E_C + q\phi_{i+1,eq}}{k_B T}\right)}\right] \quad \text{--- (C2)} \end{aligned}$$

Similar equation can be written for μ_p/D_p .

MOS capacitance in degenerate case:

Let's take a p-type substrate with non-degenerate doping. In accumulation mode, as the valence band comes closer to E_F , the acceptor impurities are partially ionized. In the inversion mode, the conduction band comes close to E_F and if there are any donor impurities, they will be partially ionized. (The overall doping is p-type, but there could be some donor impurities.) Also, we need to use the Fermi-Dirac statistics for the electrons and holes in these cases. Same arguments apply to n-type substrate. Equations derived in this Appendix can be used for calculating capacitance in these cases. The procedure is similar to what has been used for non-degenerate approximation case. We first get the equilibrium (n, p, ϕ) for a given value of gate potential using degenerate equations. We then superimpose a small ac signal. We now have additional terms involving time derivative of ϕ (see equation B3' and B4'). We also consider that the ratio D/μ depends on position with values given by equation C1 or C2.

The program MOS_capacitance.m can calculate capacitance in the degenerate case. In figure 5.1, we have shown the gate voltages where the conduction band and valence band touch the Fermi level. We definitely need to use the degenerate equations near and beyond this gate voltage. However, it turns out that the difference in capacitance is quite small whether you calculate it by degenerate or non-degenerate equations.

Appendix D: Kramer-Kronig relations

We have calculated the capacitance and conductance in two cases: diode (section 4) and MOS (section 5). If the numerical calculations are correct, the computed impedance should satisfy the Kramer-Kronig relations. The capacitance and conductance as a function of frequency for diode is plotted in fig 4.2. The numerical results fit reasonably well the theoretical equations. The admittance can be considered as a parallel combination of diffusion capacitance, diffusion conductance and junction capacitance. The diffusion admittance is given by: $Y_D = G_0(1 + j\omega\tau)$. (See equation 7.28 in Pierret, Semiconductor Device Fundamentals). It is well known that this form of impedance satisfies Kramer-Kronig relations. You can also verify this numerically. You can use MATLAB function hilbert to calculate the imaginary part of impedance from the real part (or vice-versa).

The uploaded MOS program calculates capacitance and conductance as a function of gate voltage at a fixed frequency. You can easily modify it to calculate these quantities as a function of frequency at a fixed gate voltage. The results of these calculation are shown in figure D1.

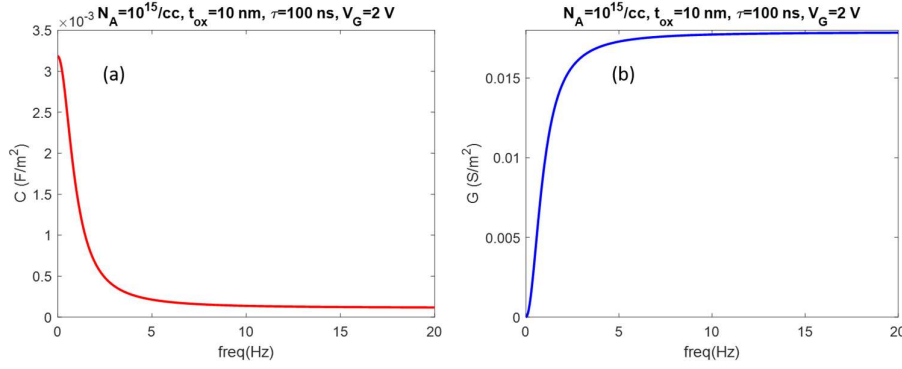


Figure D1. (a) Capacitance and (b) conductance as a function of frequency for MOS capacitor.

Here we are considering MOS as a parallel combination of conductance and capacitance. The impedance of circuit is given by: $Z(\omega) = [G(\omega) + j\omega C(\omega)]^{-1}$. This impedance diverges at $\omega=0$, as the conductance is 0 at $\omega=0$. We can therefore consider $f(\omega)=G(\omega)Z(\omega)$ as the response function. This response function corresponds to the ratio current flowing through the conductance to the applied current, and is well behaved at $\omega=0$. Figure D2 shows the real and imaginary parts of the f , and imaginary part of f calculated from the real part using Kramer-Kronig relations. MATLAB function hilbert has been used for this calculation. We can see that Kramer-Kronig relations are not satisfied at low frequencies, indicating error in the calculation. The capacitance behavior (values at $\omega=0$ and $\omega \rightarrow \infty$) as shown in figure D1.a is expected from theory. Also, the value of conductance should be 0 at $\omega=0$. As noted in figure 5.2 a and b, there is some problem in current conservation at low frequencies and MATLAB also gives warning for these calculations. It therefore seems that the calculated low frequency behaviour of capacitance and conductance is not correct.

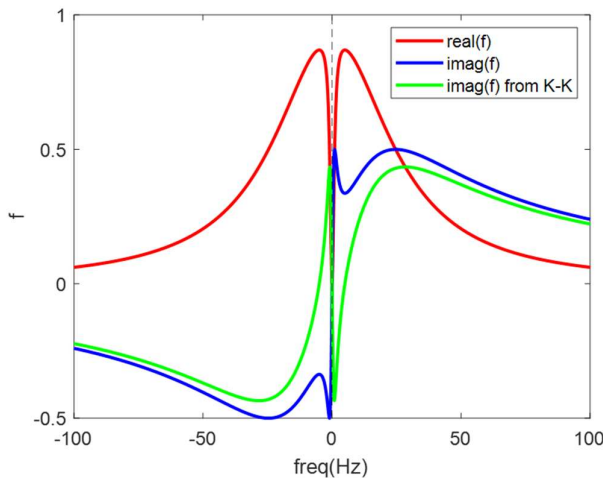


Figure D2. The real (red) and imaginary (blue) parts of the response function. The green curve is obtained from the real part using Kramer-Kronig relations.

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